Supplementary Material

Modeling Soil–Water–Atmosphere Interactions for Irrigation Scheduling and Design

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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S1. What is new in this article

The table sets each contribution against what was established before it. The derivations that follow mark which equations are taken from the literature and which are new.

Table S1: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Irrigation tools compute schedules and designs from standard procedures (Allen et al., 1998; Šimůnek et al., 2016; Steduto et al., 2009) but seldom show which relation produced a number or where it holds.	A *ledger* in which every numbered equation carries its origin (literature, fitted to data, engineered) and every number of the interface leads to one of them.	134 equations labelled: 38 literature, 18 fitted, 78 engineered.
2	Meta-models of process models speed up screening (Asher et al., 2015; Razavi et al., 2012).	A meta-model of a designed Daisy ensemble whose rank fidelity and absolute offset are both measured on held-out runs.	Spearman 0.64 to 0.97 on held-out simulations; off-axis nRMSE 6.2 % (ET) and 18.9 % (irrigation) after re-anchoring.
3	Surface, drip and sprinkler design procedures are textbook (Christiansen, 1942; Keller and Karmeli, 1974; Walker and Skogerboe, 1987).	The procedures implemented and checked against published worked examples, with three defects found and removed by that check.	Median error 0.08 %, maximum 1.81 %; a fixed advance exponent that under-predicted slow advance by up to 29 % removed; drip solver reliable to a head loss of 60 %.
4	A suction trigger is set from tables (Allen et al., 1998).	The coupled column shows how one trigger means different things in different soils.	A -300 cm trigger uses 24 to 84 % of the plant-available water; drainage ranges from 28 to 164 mm yr^{-1} under the same rain.
5	–	Two limitations of the system exposed by its own evaluation.	The leaf water potential is set by a conductivity floor; the drainage share of a mulch saving is understated in coarse soils.

#	Established before	New in this article	Evidence here
6	–	Scenario sweeps fast enough for interactive use while staying traceable.	10^5 scenarios in about 2 s on a laptop.

S2. The author's contributions: problem, derivation and pinpoints

Each contribution of the article is set out in the same way: *the problem* it answers, stated in a few lines; *the derivation*, step by step; *the result*, with the parts that matter marked by circled labels ①, ②, …; and *the pinpoints*, which say for each circled mark what it shows and why it is so. Numbers come from the data and can be recomputed in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`.

S2.1 Contribution 1: one depletion, ten soil-specific triggers

The problem

Agronomic guidance gives a depletion fraction at which to irrigate (FAO-56), while sensors read suction. A single suction trigger is therefore used on every soil. What share of the plant-available water does that trigger really represent, and which suction corresponds to the same depletion on each soil?

The derivation

(a) The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). **(b)** Because $\theta - \theta_r \propto S_e$, the depletion at suction h needs only S_e: $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$. **(c)** Set $f = f^*$ and solve for S_e^*. **(d)** Invert the van Genuchten curve.

The result

$$\underbrace{h^*}_{①} = \frac{1}{\alpha}\left(\underbrace{[S_e(100) - f^*\{S_e(100) - S_e(15\,849)\}]}_{②}^{-1/m} - 1\right)^{1/n}$$

$$\underbrace{f(300) = 0.24 \text{ to } 0.84}_{③}$$

Soil	h^* at $f^* = 0.45$ (cm)	water used at a 300 cm trigger
Sand	④ 143	84 %
Loam	284	47 %
Silty clay loam	⑤ 729	24 %

Pinpoints: what each circled mark shows and why

① The soil's own trigger: the suction at which this soil has used the chosen share of its available water.

② The effective saturation at the chosen depletion; the factor $\theta_s - \theta_r$ cancels, so only α and n are needed.

③ The same 300 cm trigger means 24 % of the water used in silty clay loam but 84 % in sand: one trigger is not one decision.

④ Sand must be irrigated at a much wetter suction because it releases most of its water at low suction.

⑤ Fine soils release water over a wide range, so the same depletion is reached at a much drier suction.

S2.2 Contribution 2: an interception store that matches the annual model

The problem

The daily model must lose to canopy interception the same water per year as the annual model, although its rain falls in random daily amounts. How large must the daily interception store be?

The result

$$\underbrace{\mathrm{E}[(R - I_c)^+]}_{①} = \mu_0 e^{-I_c/\mu_0} \;\Rightarrow\; \underbrace{r = e^{-I_c/\mu_0}}_{②} \;\Rightarrow\; \underbrace{I_c = \mu_0 \ln \frac{1}{r}}_{③}$$

Pinpoints: what each circled mark shows and why

(1) Expected throughfall of an exponential daily rain of mean μ_0 above a store I_c, obtained by integrating the exponential density.

(2) The throughfall fraction; it is set equal to the throughfall of the annual model.

(3) The store follows in closed form: with $\mu_0 = 4.5\,\text{mm}$ and $r = 0.9$, $I_c = 0.47\,\text{mm}$. The two engines therefore agree on interception by construction.

S2.3 Contribution 3: two evaporation stages joined without a jump

The result

$$Q(t) = Q_1 + \underbrace{2 e_0 \tau_E}_{(1)} \left(\sqrt{1 + \underbrace{(t - t_1)/\tau_E}_{(2)}} - 1 \right)$$

$$\underbrace{\left. \frac{\mathrm{d}Q}{\mathrm{d}t} \right|_{t_1} = e_0}_{(3)}$$

$$\underbrace{Q \propto \sqrt{t} \;\; (t - t_1 \gg \tau_E)}_{(4)}$$

Pinpoints: what each circled mark shows and why

(1) The amplitude of stage 2, chosen so that the rate is continuous.

(2) Time since the end of stage 1, in units of the transition time τ_E.

(3) At the start of stage 2 the rate equals the stage-1 rate: no artificial jump in evaporation.

(4) Long after, cumulative evaporation grows as the square root of time, the diffusion-limited law of stage 2.

S2.4 Contribution 4: a ledger of every equation

The result

Origin	Equations	Meaning
(1) Literature	38	taken from a cited source
(2) Fitted to data	18	coefficients fitted to the ensemble
(3) Engineered	78	form chosen for the software

Pinpoints: what each circled mark shows and why

(1) These carry the authority of their source and need only to be implemented correctly.

(2) These hold only within the data they were fitted to; their range is stated.

(3) These are the largest class and the ones that most need testing; labelling them tells the reader where the evidence is weakest.

S2.5 Contribution 5: a defect found and removed in surface irrigation

The problem

The surface layer under-predicted slow advance (up to $-29\,\%$ in a published example). Why?

The result

$$\underbrace{x(t)}_{(1)} = \frac{Qt}{\sigma_y A_0 + \underbrace{\sigma_z(r)}_{(2)}\, k t^{A} w}, \qquad \sigma_z = \frac{A + \underbrace{r}_{(3)}(1-A) + 1}{(1+A)(1+r)}$$

Pinpoints: what each circled mark shows and why

(1) The advance distance at time t.

(2) The subsurface shape factor depends on the advance exponent r.

(3) The first release fixed r; for slow advance the true r is smaller, σ_z too small and the advance too short. Iterating r with the computed advance removed the error, now within -0.2 to $+0.5\,\%$ of the published advance times.

S3. A step-by-step guide for students

Each step explains one engine of the article with a worked example; the numbers can be reproduced in sheet `EQ_P2` of the workbook.

S3.1 Step 1: from a suction trigger to a share of the available water

The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). The depletion at a suction h is $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$. *Example (loam, $\alpha = 0.036$, $n = 1.56$, $m = 0.359$).* $S_e(100) = 0.466$, $S_e(15\,849) = 0.029$, $S_e(300) = 0.262$; $f(300) = (0.466 - 0.262)/(0.466 - 0.029) = 0.47$: at a 300 cm trigger the loam has used 47 % of its available water. In sand the same trigger means 84 %, in silty clay loam 24 %.

S3.2 Step 2: from a chosen depletion to a soil's own trigger

Choose $f^* = 0.45$. Then $S_e^* = S_e(100) - f^*[S_e(100) - S_e(15\,849)]$ and the trigger is $h^* = \alpha^{-1}(S_e^{*\,-1/m} - 1)^{1/n}$. *Example (loam).* $S_e^* = 0.466 - 0.45 \times 0.437 = 0.269$; $0.269^{-1/0.359} = 38.8$; $38.8 - 1 = 37.8$; $37.8^{1/1.56} = 10.2$; $10.2/0.036 = 284$ cm.

S3.3 Step 3: interception by the canopy

If daily rain is exponential with mean μ_0, a store I_c lets through the fraction $r = e^{-I_c/\mu_0}$, so $I_c = \mu_0 \ln(1/r)$. *Example:* $\mu_0 = 4.5$ mm and $r = 0.9$ give $I_c = 4.5 \times 0.105 = 0.47$ mm.

S3.4 Step 4: infiltration (Green–Ampt)

Water enters behind a sharp front. With K_s, front suction ψ_f and water-content jump $\Delta\theta$, the infiltrated depth F after time t solves $F - \psi_f \Delta\theta \ln(1 + F/(\psi_f \Delta\theta)) = K_s t$, found by trying values of F until both sides agree.

S3.5 Step 5: surface irrigation advance

The water admitted ($Q\,t$) is split between the surface and the soil; dividing by the volume per metre gives the advance $x(t) = Qt/(\sigma_y A_0 + \sigma_z k t^A w)$. The efficiencies follow from the stored, percolated and runoff volumes and add to one.

S3.6 Step 6: friction in a drip lateral

The Hazen–Williams loss of the full flow is reduced by the Christiansen factor because the flow falls along the lateral. *Example:* 50 outlets give $F_c = 1/2.852 + 1/100 + \sqrt{0.852}/15\,000 = 0.361$; a 100 m lateral of 16 mm with 0.5 L s^{-1} at the inlet loses about 17.6 m.

S3.7 Step 7: a sprinkler nozzle

$q = 0.95 A_n \sqrt{2 g h_n}$. *Example:* a 4 mm nozzle at 30 m head: $A_n = 1.26 \times 10^{-5}$ m^2, $\sqrt{2 \times 9.81 \times 30} = 24.3$ m s^{-1}, $q = 2.9 \times 10^{-4}$ m^3 s^{-1} = 17.4 L min^{-1}.

S3.8 Step 8: reproduce everything

Sheet `EQ_P2` holds Steps 1, 2, 3, 6 and 7 as formulas; the verification scripts of the article are listed in its last section.

S4. Derivations of the equations

Most derivations of Section S2 are of established relations, given so that every equation of the article can be followed from first principles: the capillary law, the Kelvin relation, Green–Ampt infiltration, the volume balance of surface irrigation, Hazen–Williams losses with the Christiansen factor, emission uniformity and nozzle discharge (Christiansen, 1942; Green and Ampt, 1911; Keller and Karmeli, 1974; Walker and Skogerboe, 1987). New in this article are the inversion of the depletion fraction through the van Genuchten curve into a soil-specific trigger (S2.3), the smooth joining of the two evaporation stages (S2.5), and the choice of the interception store that makes the daily model reproduce the annual interception (S2.8); the contribution of the article is the system that joins these relations, its ledger and its verification (Table S1).

S4.1 Water balance of the column

Mass conservation of the column between the surface and 1 m gives $P + I = T + E + E_{\mathrm{oth}} + D + \Delta S$ per unit area, with all terms in mm; every engine is constructed so that this balance closes, and its residual is the test of Engine 2.

S4.2 Capillary radius

The pressure jump across a meniscus in a cylindrical pore of radius r is $\Delta p = 2\sigma\cos\gamma_c/r$ (Young–Laplace). Equating with the suction head $\rho_w g h$ gives $r_c = 2\sigma\cos\gamma_c/(\rho_w g h)$; with $\sigma = 0.0728\,\mathrm{N\,m^{-1}}$, $\gamma_c = 0$, $\rho_w = 1000\,\mathrm{kg\,m^{-3}}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$, $r_c = 1.49\times10^{-5}\,\mathrm{m^2}/h = 0.149\,\mathrm{cm^2}/h$: the pores emptied at a suction h are those wider than r_c.

S4.3 Depletion and the trigger law

The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). The depletion at suction h is the share of that water that has been used, $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$, because $\theta - \theta_r$ is proportional to S_e and the factor $\theta_s - \theta_r$ cancels. The suction that corresponds to a chosen depletion f^* is found by inverting: $S_e(h^*) = S_e(100) - f^*[S_e(100) - S_e(15\,849)]$ and $h^* = \alpha^{-1}(S_e^{-1/m} - 1)^{1/n}$. A soil-independent depletion therefore gives a soil-specific suction, which is why one suction trigger uses different shares of the available water in different soils.

S4.4 Consolidation of the tilled layer

If the excess pore space decays at a rate proportional to itself, $\mathrm{d}p/\mathrm{d}t = -\tau_c(p - p_c)$, then $p(t) = p_c + (p_0 - p_c)e^{-\tau_c t}$, with p_0 set by the tillage and p_c the consolidated state.

S4.5 Two-stage evaporation

In stage 1 the surface is wet and evaporation runs at the demand e_0, so $Q = q_0 + e_0 t$ until the cumulative stage-1 limit Q_1 is reached at $t_1 = (Q_1 - q_0)/e_0$. In stage 2 evaporation is limited by diffusion through the dry layer; for a desorptivity-controlled flux $e = S/(2\sqrt{t})$ the cumulative loss grows as $\sqrt{t}$. The form of the paper, $Q = Q_1 + 2e_0\tau_E(\sqrt{1 + (t - t_1)/\tau_E} - 1)$, starts stage 2 at the stage-1 rate (its derivative at $t = t_1$ is e_0) and tends to the $\sqrt{t}$ law for $t - t_1 \gg \tau_E$, which joins the two stages smoothly (Philip, 1957; Ritchie, 1972).

S4.6 Water potential of the air

For water vapour in equilibrium with liquid water, the chemical potential difference per unit volume is $\psi = (\mathcal{R}T_K/V_w)\ln\mathrm{RH}$ (Kelvin). At 20 °C, $\mathcal{R}T_K/V_w = 8.314\times293.15/1.8\times10^{-5} =$

1.353×10^8 Pa, that is 135.3 MPa, so $\psi_{\text{air}} \simeq 135.3 \ln \text{RH}$ MPa.

S4.7 The soil–plant–atmosphere path

With steady flow q through soil, root, xylem and leaf in series, each segment drops the potential by qR_j (the van den Honert analogue of Ohm's law), so $\psi_{\text{leaf}} = \psi_{\text{RZ}} - q(R_{\text{soil}} + R_{\text{root}} + R_{\text{xyl}} + R_{\text{leaf}})$. The soil resistance grows as the conductivity falls; bounding K_r from below (the floor of the paper) bounds R_{soil} and therefore the leaf potential, which is why that potential is set by the floor rather than by the soil when the soil is dry.

S4.8 Rainfall and interception of the daily model

If daily rain depths are exponential with mean μ_0, the expected throughfall above an interception store I_c is $\mathrm{E}[(R - I_c)^+] = \int_{I_c}^{\infty} (R - I_c)\mu_0^{-1} e^{-R/\mu_0} \mathrm{d}R = \mu_0 e^{-I_c/\mu_0}$. The throughfall fraction is $r = e^{-I_c/\mu_0}$, hence $I_c = \mu_0 \ln(1/r)$: the store is chosen so that the expected interception equals the annual interception of the annual model.

S4.9 Green–Ampt infiltration

Behind a sharp wetting front at depth L_f with uniform water-content jump $\Delta\theta$ and front suction ψ_f, Darcy's law gives the infiltration rate $f = K_s(L_f + \psi_f)/L_f$ (ponding head neglected). With $F = L_f \Delta\theta$, $f = \mathrm{d}F/\mathrm{d}t = K_s(1 + \psi_f \Delta\theta / F)$. Separating variables and integrating from $F = 0$ at $t = 0$:

$$F - \psi_f \Delta\theta \ln\Big(1 + \frac{F}{\psi_f \Delta\theta}\Big) = K_s t. \tag{S1}$$

S4.10 Surface irrigation: volume balance and advance

At time t after the start, the volume admitted, Qt, is stored on the surface (mean flow area $\sigma_y A_0$ over the advance length x) and in the soil (mean infiltrated depth $\sigma_z Z(t)$ over the wetted width w): $Qt = \sigma_y A_0 x + \sigma_z k t^A w x$, so $x(t) = Qt/(\sigma_y A_0 + \sigma_z k t^A w)$. If the advance follows a power law $x \propto t^r$ and the intake opportunity time at distance s is $t - t_x(s)$, the mean of Z along the advance is $Z(t)$ times

$$\sigma_z = \int_0^1 (1 - \xi^{1/r})^A \mathrm{d}\xi \;\simeq\; \frac{A + r(1 - A) + 1}{(1 + A)(1 + r)}, \tag{S2}$$

where the right-hand side is the closed approximation used in surface-irrigation practice (Walker and Skogerboe, 1987); it is exact for $A = 0$ and $A = 1$. The volumes applied, required, infiltrated, stored, percolated and run off satisfy $V_{\mathrm{app}} = V_{\mathrm{inf}} + V_{\mathrm{ro}}$ and $V_{\mathrm{inf}} = V_{\mathrm{st}} + V_{\mathrm{dp}}$, so the application efficiency, deep percolation and runoff fractions add to one.

S4.11 Drip laterals

The Hazen–Williams friction loss over a pipe segment of length Δx carrying Q_j is $h_{f,j} = 10.67 \Delta x\, Q_j^{1.852} / (C^{1.852} D^{4.871})$. Along a lateral with N equal outlets the flow falls linearly towards the end; summing the losses of the segments, $h_f = \frac{10.67 L Q_0^{1.852}}{C^{1.852} D^{4.871}} \frac{1}{N^{2.852}} \sum_{j=1}^{N} j^{1.852}$. The Euler–Maclaurin formula gives $\sum_{j=1}^{N} j^m \simeq \frac{N^{m+1}}{m+1} + \frac{N^m}{2} + \ldots$ with $m = 1.852$, so the loss is the full-flow loss times $F_c = \frac{1}{m+1} + \frac{1}{2N} + \ldots$; Christiansen (1942) approximated the remaining terms by $\sqrt{m-1}/(6N^2)$, which gives the factor of the paper, $F_c = \frac{1}{2.852} + \frac{1}{2N} + \frac{\sqrt{0.852}}{6N^2}$. Each emitter discharges $q_j = q(h_j/H_0)^{x_e}$, which couples the flows to the heads and is solved by iteration.

S4.12 Emission uniformity

For emitters with manufacturing variation c_v and n_e emitters per plant, the lower quarter of plant flows lies about $1.27 c_v / \sqrt{n_e}$ below the mean (normal distribution), and the hydraulic variation adds the ratio $q_{\mathrm{min}}/\bar{q}$; their product is the design emission uniformity of Keller and Karmeli (1974).

S4.13 Sprinkler discharge

Bernoulli's equation across a nozzle gives the jet speed $v = \sqrt{2 g h_n}$, and the discharge is $q = C_d A_n \sqrt{2 g h_n}$ with $C_d \approx 0.95$; the factor 6×10^4 converts $\mathrm{m^3\,s^{-1}}$ to $\mathrm{L\,min^{-1}}$. The Christiansen coefficient of uniformity is $CU = 1 - \sum |z_i - \bar{z}| / (n \bar{z})$.

S5. Why the results happen

Why the same trigger uses different shares of the available water. A suction trigger marks a point on the retention curve. In a sand most available water is released at low suctions, so a -300 cm trigger is reached late (much of the water used); in a clay the water is released over a wide range, so the same suction is reached early.

Why drainage differs under the same rain. Drainage is the water that exceeds what the profile holds and the crop uses; profiles that hold less (sands) and conduct more pass more water below 1 m.

Why stage-1 evaporation lasts longer in fine soils. Fine soils hold more water near the surface and conduct it upward over longer distances, so the surface stays wet longer.

Why the leaf water potential is set by a conductivity floor. The series resistance of the soil grows without bound as the soil dries; the software bounds the relative conductivity from below to keep the calculation finite, and in dry soil that bound, not the soil, then decides the leaf potential.

Why the drainage share of a mulch saving is understated in coarse soils. The engines split the saving with ratios that do not contain the soil's drainage competition; in coarse soils, where most saved water drains, that split gives too little to drainage.

Why the meta-model ranks well but has an offset. It is built from ratios of simulated outcomes, which preserve the order of the scenarios, while its absolute level is anchored to one reference run, so an error in the anchor shifts all values.

Why the daily model fails in sand at dry triggers. In sand dry triggers lie beyond the suction the crop can produce in the soil, so the full simulation seldom irrigates, while the daily bucket, which has no conductivity limit, still reaches the trigger.

S6. The data

Table S2 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

Table S2: The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

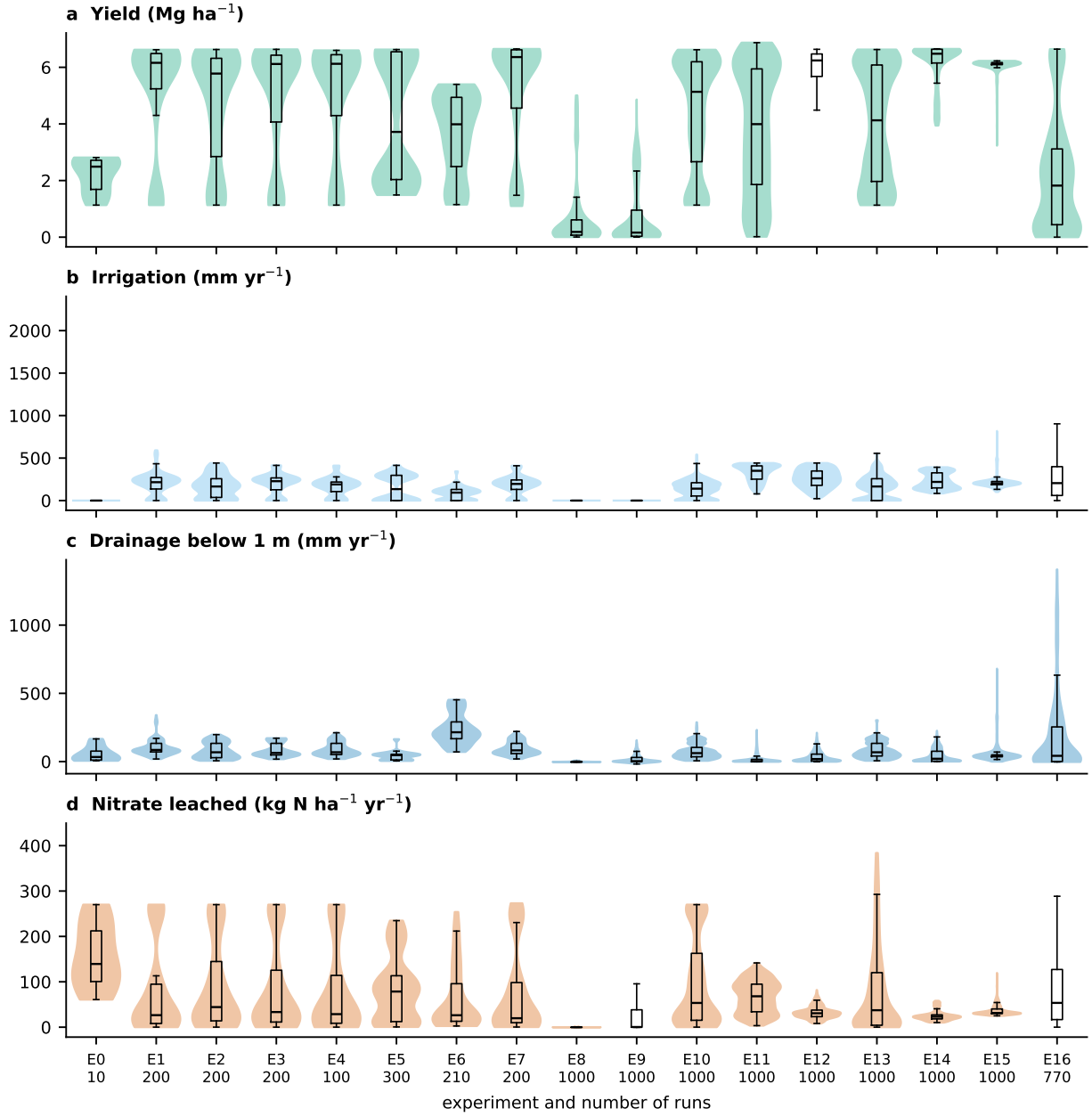


Figure S1: Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

p3_swa_vars.json

Source: `Manuscripts/results/p3_swa_vars.json`; 650 values, first 220 shown. Workbook sheet `R_p3_swa_vars`.

quantity	value
[0].soil	S01_CoarseLoam
[0].label	Coarse Loam
[0].hyd.alpha	0.09613
[0].hyd.n	1.353
[0].hyd.sand	45

quantity	value
[0].hyd.silt	35
[0].hyd.clay	20
[0].hyd.thetaS	0.4235
[0].hyd.thetaR	0.022
[0].hyd.AWC	1.5
[0].hyd.f300	0.3795
[0].hyd.hStar	387.3
[0].hyd.Se100	0.4443
[0].e1.evap	174.6
[0].e1.transp	239
[0].e1.et	720.3
[0].e1.irr	282.2
[0].e1.yield	6.627
[0].e1.drain	27.62
[0].e1.nl	14.56
[0].e1.rain	464
[0].e1.other	306.7
[0].e1.store	-1.67
[0].e1.ie	2.348
[0].e2.Rmax	93.58
[0].e2.meanRootMax	89.94
[0].e2.z50	19.25
[0].e2.z90	47
[0].e2.kt	1.641
[0].e2.dose	28.22
[0].e2.fTrig	0.3795
[0].e2.irrMean	283.7
[0].e2.eventsPerYear	10.05
[0].e2.uptakeProfile	30.27, 24.66, 23.52, 22.37, 22.36, 21.16, 19.86, 18.68, 20.95, 14.61, 13.69, 12.82 ... (40 values)
[0].e2.sProfile	0.4658, 0.5419, 0.6077, 0.6541, 0.7093, 0.7365, 0.7559, 0.7728, 0.7867, 0.7988, 0.81, 0.821 ... (40 values)
[0].e2.wetFrontIrr	45
[0].e2.drainUnscaled	7.334
[0].e2.interceptionUnscaled	312.7
[0].e2.evapUnscaled	117
[0].e2.transpUnscaled	335.7
[0].spac.psiRZ	-0.0494
[0].spac.psiRoot	-1.814
[0].spac.psiLeaf	-2.243
[0].spac.Kr	8.391e-06
[0].spac.closedShare	0.7405
[0].spac.psiAir	-56.3
[0].spac.days	1715
[0].evap.e0	4.2
[0].evap.Q1	5.991
[0].evap.tau	0.0225
[0].evap.t1	1.426

quantity	value
[0].evap.lost12	9.903
[0].evap.dtheta	0.3514
[0].evap.hi	2.601
[0].evap.hs	41.61
[0].mulch.dE	127.4
[0].mulch.dI	34.86
[0].mulch.dD	7.26
[0].mulch.dT	17.84
[0].mulch.fI	0.2736
[0].mulch.fD	0.05697
[0].mulch.fT	0.14
[0].mulch.fRest	0.5294
[0].regime	0.7268
[0].kappaN	0.3783
[1].soil	S02_CoarseSand
[1].label	Coarse Sand
[1].hyd.alpha	0.05358
[1].hyd.n	1.6
[1].hyd.sand	88
[1].hyd.silt	8
[1].hyd.clay	4
[1].hyd.thetaS	0.3736
[1].hyd.thetaR	0.0124
[1].hyd.AWC	0.5
[1].hyd.f300	0.4964
[1].hyd.hStar	261.9
[1].hyd.Se100	0.3563
[1].e1.evap	117.8
[1].e1.transp	236.7
[1].e1.et	716.3
[1].e1.irr	413.8
[1].e1.yield	6.449
[1].e1.drain	163.6
[1].e1.nl	62.55
[1].e1.rain	464
[1].e1.other	361.8
[1].e1.store	-2.18
[1].e1.ie	1.559
[1].e2.Rmax	100
[1].e2.meanRootMax	96.15
[1].e2.z50	24.5
[1].e2.z90	64.75
[1].e2.kt	1.12
[1].e2.dose	35
[1].e2.fTrig	0.4964
[1].e2.irrMean	423.5
[1].e2.eventsPerYear	12.1

quantity	value
[1].e2.uptakeProfile	8.878, 10.93, 12.22, 12.53, 13.49, 12.93, 12.01, 11.32, 11.73, 9.564, 8.149, 7.661 ... (40 values)
[1].e2.sProfile	0.2935, 0.4015, 0.4911, 0.552, 0.6278, 0.6603, 0.6845, 0.7038, 0.7202, 0.7356, 0.7505, 0.7642 ... (40 values)
[1].e2.wetFrontIrr	100
[1].e2.drainUnscaled	259
[1].e2.interceptionUnscaled	369.4
[1].e2.evapUnscaled	61.06
[1].e2.transpUnscaled	219.3
[1].spac.psiRZ	-0.03899
[1].spac.psiRoot	-1.848
[1].spac.psiLeaf	-2.275
[1].spac.Kr	5.581e-05
[1].spac.closedShare	0.7504
[1].spac.psiAir	-56.3
[1].spac.days	1715
[1].evap.e0	4.2
[1].evap.Q1	7.849
[1].evap.tau	0.0096
[1].evap.t1	1.869
[1].evap.lost12	10.39
[1].evap.dtheta	0.3392
[1].evap.hi	4.666
[1].evap.hs	74.66
[1].mulch.dE	102.5
[1].mulch.dI	51.11
[1].mulch.dD	7.26
[1].mulch.dT	14.76
[1].mulch.fI	0.4988
[1].mulch.fD	0.07086
[1].mulch.fT	0.1441
[1].mulch.fRest	0.2862
[1].regime	-0.1628
[1].kappaN	0.3783
[2].soil	S03_Sand
[2].label	Sand
[2].hyd.alpha	0.145
[2].hyd.n	2.68
[2].hyd.sand	92
[2].hyd.silt	5
[2].hyd.clay	3
[2].hyd.thetaS	0.3693
[2].hyd.thetaR	0.0118
[2].hyd.AWC	0.6
[2].hyd.f300	0.8422
[2].hyd.hStar	142.8
[2].hyd.Se100	0.01119
[2].e1.evap	58.2

quantity	value
[2].e1.transp	202.3
[2].e1.et	669
[2].e1.irr	278.2
[2].e1.yield	4.984
[2].e1.drain	76.52
[2].e1.nl	85.16
[2].e1.rain	464
[2].e1.other	408.5
[2].e1.store	-3.32
[2].e1.ie	1.792
[2].e2.Rmax	100
[2].e2.meanRootMax	95.84
[2].e2.z50	27
[2].e2.z90	52.38
[2].e2.kt	2.045
[2].e2.dose	27.82
[2].e2.fTrig	0.8422
[2].e2.irrMean	285.1
[2].e2.eventsPerYear	10.25
[2].e2.uptakeProfile	14.55, 15.46, 16.02, 15.59, 16.18, 15.72, 15.22, 14.97, 15.88, 14.28, 13.66, 13.55 ... (40 values)
[2].e2.sProfile	0.2917, 0.338, 0.3738, 0.396, 0.4246, 0.4392, 0.4515, 0.4625, 0.4765, 0.4922, 0.5086, 0.5228 ... (40 values)
[2].e2.wetFrontIrr	55
[2].e2.drainUnscaled	5.15
[2].e2.interceptionUnscaled	424.5
[2].e2.evapUnscaled	32.31
[2].e2.transpUnscaled	314.8
[2].spac.psiRZ	-0.02021
[2].spac.psiRoot	-1.694
[2].spac.psiLeaf	-2.114
[2].spac.Kr	2.075e-09
[2].spac.closedShare	0.712
[2].spac.psiAir	-56.3
[2].spac.days	1715
[2].evap.e0	4.2
[2].evap.Q1	12.71
[2].evap.tau	0.0084
[2].evap.t1	3.027
[2].evap.lost12	14.95
[2].evap.dtheta	0.3521
[2].evap.hi	1.724
[2].evap.hs	27.59
[2].mulch.dE	49.62
[2].mulch.dI	34.36
[2].mulch.dD	7.26
[2].mulch.dT	15.1
[2].mulch.fI	0.6924

quantity	value
[2].mulch.fD	0.1463
[2].mulch.fT	0.3043
[2].mulch.fRest	0
[2].regime	-0.136
[2].kappaN	0.3783
[3].soil	S04_LoamySand
[3].label	Loamy Sand
[3].hyd.alpha	0.124
[3].hyd.n	2.28
[3].hyd.sand	82
[3].hyd.silt	12
[3].hyd.clay	6
[3].hyd.thetaS	0.3804
[3].hyd.thetaR	0.0136
[3].hyd.AWC	0.9
[3].hyd.f300	0.7557
[3].hyd.hStar	159.5
[3].hyd.Se100	0.03978
[3].e1.evap	73.95
[3].e1.transp	215
[3].e1.et	691.2
[3].e1.irr	293
[3].e1.yield	5.456
[3].e1.drain	70.14
[3].e1.nl	61.13
[3].e1.rain	464
[3].e1.other	402.2
[3].e1.store	-4.31
[3].e1.ie	1.862
[3].e2.Rmax	100

p3_e1_sens.json

Source: `Manuscripts/results/p3_e1_sens.json`; 223 values, first 220 shown. Workbook sheet `R_p3_e1_sens`.

quantity	value
base.irr	282.2
base.et	720.3
base.drain	27.62
base.yield	6.627
base.nl	14.56
base.evap	174.6
base.transp	239
F[0].name	Rainfall scale

quantity	value
F[0].labLo	x0.7
F[0].labHi	x1.3
F[0].lo.irr	430.2
F[0].lo.et	725.9
F[0].lo.drain	10.64
F[0].lo.yield	3.655
F[0].lo.nl	10.5
F[0].lo.evap	124.6
F[0].lo.transp	249
F[0].hi.irr	188.7
F[0].hi.et	719.2
F[0].hi.drain	87.7
F[0].hi.yield	6.633
F[0].hi.nl	17.31
F[0].hi.evap	199.9
F[0].hi.transp	235.4
F[1].name	Warming
F[1].labLo	-1 C
F[1].labHi	+2 C
F[1].lo.irr	224.1
F[1].lo.et	643.4
F[1].lo.drain	35.81
F[1].lo.yield	6.633
F[1].lo.nl	6.34
F[1].lo.evap	178.2
F[1].lo.transp	218.3
F[1].hi.irr	314.3
F[1].hi.et	759.8
F[1].hi.drain	26.06
F[1].hi.yield	2.084
F[1].hi.nl	100.5
F[1].hi.evap	178.8
F[1].hi.transp	171.9
F[2].name	Trigger suction
F[2].labLo	-150 cm
F[2].labHi	-1000 cm
F[2].lo.irr	334.6
F[2].lo.et	747.1
F[2].lo.drain	43.59
F[2].lo.yield	6.633
F[2].lo.nl	13.14
F[2].lo.evap	174.8
F[2].lo.transp	244.4
F[2].hi.irr	208.4
F[2].hi.et	671.7
F[2].hi.drain	15.74
F[2].hi.yield	6.379
F[2].hi.nl	18.33

quantity	value
F[2].hi.evap	164.7
F[2].hi.transp	230.8
F[3].name	Sensor depth
F[3].labLo	10 cm
F[3].labHi	40 cm
F[3].lo.irr	315.1
F[3].lo.et	751.1
F[3].lo.drain	26.24
F[3].lo.yield	6.633
F[3].lo.nl	1.89
F[3].lo.evap	172.1
F[3].lo.transp	255.8
F[3].hi.irr	269.6
F[3].hi.et	709.6
F[3].hi.drain	28.63
F[3].hi.yield	6.583
F[3].hi.nl	18.36
F[3].hi.evap	173.3
F[3].hi.transp	234.2
F[4].name	Mulch vapour-flux factor
F[4].labLo	0.4
F[4].labHi	0.7
F[4].lo.irr	217.7
F[4].lo.et	639.3
F[4].lo.drain	50.51
F[4].lo.yield	6.633
F[4].lo.nl	4.66
F[4].lo.evap	76.19
F[4].lo.transp	257.2
F[4].hi.irr	257.8
F[4].hi.et	690
F[4].hi.drain	36.01
F[4].hi.yield	6.633
F[4].hi.nl	8.87
F[4].hi.evap	115.6
F[4].hi.transp	247.7
F[5].name	Irrigation method
F[5].labLo	overhead
F[5].labHi	drip 20-30 cm
F[5].lo.irr	281.9
F[5].lo.et	718.4
F[5].lo.drain	28.73
F[5].lo.yield	6.633
F[5].lo.nl	14.14
F[5].lo.evap	186.9
F[5].lo.transp	140.9
F[5].hi.irr	200.7
F[5].hi.et	645

quantity	value
F[5].hi.drain	31.14
F[5].hi.yield	6.564
F[5].hi.nl	15.87
F[5].hi.evap	183.3
F[5].hi.transp	258
F[6].name	Tillage
F[6].labLo	no-till
F[6].labHi	conv (base)
F[6].lo.irr	276
F[6].lo.et	720.4
F[6].lo.drain	21.35
F[6].lo.yield	6.626
F[6].lo.nl	12.95
F[6].lo.evap	178.8
F[6].lo.transp	239
F[6].hi.irr	282.2
F[6].hi.et	720.3
F[6].hi.drain	27.62
F[6].hi.yield	6.627
F[6].hi.nl	14.56
F[6].hi.evap	174.6
F[6].hi.transp	239
F[7].name	Nitrogen rate
F[7].labLo	55%
F[7].labHi	130%
F[7].lo.irr	282.2
F[7].lo.et	720.3
F[7].lo.drain	27.62
F[7].lo.yield	6.372
F[7].lo.nl	0
F[7].lo.evap	174.6
F[7].lo.transp	239
F[7].hi.irr	282.2
F[7].hi.et	720.3
F[7].hi.drain	27.62
F[7].hi.yield	6.627
F[7].hi.nl	70.09
F[7].hi.evap	174.6
F[7].hi.transp	239
SO[0].soil	S01_CoarseLoam
SO[0].irr	282.2
SO[0].et	720.3
SO[0].drain	27.62
SO[0].yield	6.627
SO[0].nl	14.56
SO[0].evap	174.6
SO[0].transp	239
SO[1].soil	S02_CoarseSand

quantity	value
SO[1].irr	413.8
SO[1].et	716.3
SO[1].drain	163.6
SO[1].yield	6.449
SO[1].nl	62.55
SO[1].evap	117.8
SO[1].transp	236.7
SO[2].soil	S03_Sand
SO[2].irr	278.2
SO[2].et	669
SO[2].drain	76.52
SO[2].yield	4.984
SO[2].nl	85.16
SO[2].evap	58.2
SO[2].transp	202.3
SO[3].soil	S04_LoamySand
SO[3].irr	293
SO[3].et	691.2
SO[3].drain	70.14
SO[3].yield	5.456
SO[3].nl	61.13
SO[3].evap	73.95
SO[3].transp	215
SO[4].soil	S05_SandyLoam
SO[4].irr	298.3
SO[4].et	710
SO[4].drain	57.78
SO[4].yield	6.198
SO[4].nl	30.79
SO[4].evap	102.5
SO[4].transp	229.8
SO[5].soil	S06_Loam
SO[5].irr	301.6
SO[5].et	719.8
SO[5].drain	50.36
SO[5].yield	6.585
SO[5].nl	10.12
SO[5].evap	137.8
SO[5].transp	238.4
SO[6].soil	S07_SiltLoam
SO[6].irr	305.4
SO[6].et	720.5
SO[6].drain	52.15
SO[6].yield	6.619
SO[6].nl	7.85
SO[6].evap	159.5
SO[6].transp	239.2
SO[7].soil	S08_SandyClayLoam

quantity	value
SO[7].irr	298.9
SO[7].et	718.6
SO[7].drain	49.98
SO[7].yield	6.539
SO[7].nl	9.03
SO[7].evap	123.9
SO[7].transp	237.2
SO[8].soil	S09_ClayLoam
SO[8].irr	305.6
SO[8].et	720.5
SO[8].drain	54.02
SO[8].yield	6.623
SO[8].nl	4.42
SO[8].evap	155.3
SO[8].transp	239.2
SO[9].soil	S10_SiltyClayLoam
SO[9].irr	303.2
SO[9].et	720.5
SO[9].drain	51.68
SO[9].yield	6.624

p3_l_sens.json

Source: `Manuscripts/results/p3_l_sens.json`; 649 values, first 220 shown. Workbook sheet `R_p3_l_sens`.

quantity	value
sprKeys	T, arr, ar, gross, I, ro, st, ex, lq, Ea, Es, DU ... (32 values)
spr[0].P	200
spr[0].u	0
spr[0].CU	0.8197
spr[0].Ea	0.7509
spr[0].drift	None
spr[0].keys	None
spr[1].P	200
spr[1].u	1
spr[1].CU	0.82
spr[1].Ea	0.7478
spr[1].drift	None
spr[1].keys	None
spr[2].P	200
spr[2].u	2
spr[2].CU	0.8211
spr[2].Ea	0.7451
spr[2].drift	None

quantity	value
spr[2].keys	None
spr[3].P	200
spr[3].u	3
spr[3].CU	0.822
spr[3].Ea	0.7416
spr[3].drift	None
spr[3].keys	None
spr[4].P	200
spr[4].u	4
spr[4].CU	0.823
spr[4].Ea	0.7375
spr[4].drift	None
spr[4].keys	None
spr[5].P	200
spr[5].u	5
spr[5].CU	0.8246
spr[5].Ea	0.7333
spr[5].drift	None
spr[5].keys	None
spr[6].P	300
spr[6].u	0
spr[6].CU	0.962
spr[6].Ea	0.9163
spr[6].drift	None
spr[6].keys	None
spr[7].P	300
spr[7].u	1
spr[7].CU	0.962
spr[7].Ea	0.9103
spr[7].drift	None
spr[7].keys	None
spr[8].P	300
spr[8].u	2
spr[8].CU	0.9622
spr[8].Ea	0.9039
spr[8].drift	None
spr[8].keys	None
spr[9].P	300
spr[9].u	3
spr[9].CU	0.9624
spr[9].Ea	0.8964
spr[9].drift	None
spr[9].keys	None
spr[10].P	300
spr[10].u	4
spr[10].CU	0.9628
spr[10].Ea	0.8881
spr[10].drift	None

quantity	value
spr[10].keys	None
spr[11].P	300
spr[11].u	5
spr[11].CU	0.9634
spr[11].Ea	0.8791
spr[11].drift	None
spr[11].keys	None
spr[12].P	400
spr[12].u	0
spr[12].CU	0.957
spr[12].Ea	0.8892
spr[12].drift	None
spr[12].keys	None
spr[13].P	400
spr[13].u	1
spr[13].CU	0.9569
spr[13].Ea	0.8821
spr[13].drift	None
spr[13].keys	None
spr[14].P	400
spr[14].u	2
spr[14].CU	0.9568
spr[14].Ea	0.8738
spr[14].drift	None
spr[14].keys	None
spr[15].P	400
spr[15].u	3
spr[15].CU	0.9567
spr[15].Ea	0.8644
spr[15].drift	None
spr[15].keys	None
spr[16].P	400
spr[16].u	4
spr[16].CU	0.9567
spr[16].Ea	0.8547
spr[16].drift	None
spr[16].keys	None
spr[17].P	400
spr[17].u	5
spr[17].CU	0.9569
spr[17].Ea	0.8446
spr[17].drift	None
spr[17].keys	None
drip[0].D	13
drip[0].LL	20
drip[0].Eu	0.9104
drip[0].hf	0.05446
drip[1].D	13

quantity	value
drip[1].LL	40
drip[1].Eu	0.9067
drip[1].hf	0.3546
drip[2].D	13
drip[2].LL	60
drip[2].Eu	0.8972
drip[2].hf	1.069
drip[3].D	13
drip[3].LL	80
drip[3].Eu	0.8802
drip[3].hf	2.193
drip[4].D	13
drip[4].LL	100
drip[4].Eu	0.8556
drip[4].hf	3.519
drip[5].D	13
drip[5].LL	120
drip[5].Eu	0.8212
drip[5].hf	4.934
drip[6].D	13
drip[6].LL	140
drip[6].Eu	0.7786
drip[6].hf	6.208
drip[7].D	13
drip[7].LL	160
drip[7].Eu	0.7313
drip[7].hf	7.218
drip[8].D	13
drip[8].LL	180
drip[8].Eu	0.6773
drip[8].hf	8.028
drip[9].D	13
drip[9].LL	200
drip[9].Eu	0.604
drip[9].hf	9.5
drip[10].D	13
drip[10].LL	220
drip[10].Eu	0.611
drip[10].hf	9.5
drip[11].D	13
drip[11].LL	240
drip[11].Eu	0.6272
drip[11].hf	8.392
drip[12].D	13
drip[12].LL	260
drip[12].Eu	0.5585
drip[12].hf	9.082
drip[13].D	13

quantity	value
drip[13].LL	280
drip[13].Eu	0.511
drip[13].hf	9.5
drip[14].D	13
drip[14].LL	300
drip[14].Eu	0.5431
drip[14].hf	9.5
drip[15].D	16
drip[15].LL	20
drip[15].Eu	0.9109
drip[15].hf	0.01986
drip[16].D	16
drip[16].LL	40
drip[16].Eu	0.9095
drip[16].hf	0.1314
drip[17].D	16
drip[17].LL	60
drip[17].Eu	0.906
drip[17].hf	0.4117
drip[18].D	16
drip[18].LL	80
drip[18].Eu	0.8996
drip[18].hf	0.9007
drip[19].D	16
drip[19].LL	100
drip[19].Eu	0.8901
drip[19].hf	1.572
drip[20].D	16
drip[20].LL	120
drip[20].Eu	0.8762
drip[20].hf	2.44
drip[21].D	16
drip[21].LL	140
drip[21].Eu	0.8579
drip[21].hf	3.414
drip[22].D	16
drip[22].LL	160
drip[22].Eu	0.8361
drip[22].hf	4.379
drip[23].D	16
drip[23].LL	180
drip[23].Eu	0.8093
drip[23].hf	5.341
drip[24].D	16
drip[24].LL	200
drip[24].Eu	0.7787
drip[24].hf	6.214
drip[25].D	16

quantity	value
drip[25].LL	220
drip[25].Eu	0.746
drip[25].hf	6.946
drip[26].D	16
drip[26].LL	240
drip[26].Eu	0.7094
drip[26].hf	7.586
drip[27].D	16
drip[27].LL	260
drip[27].Eu	0.6708

p3_e2_closure.json

Source: `Manuscripts/results/p3_e2_closure.json`; 900 values, first 220 shown. Workbook sheet `R_p3_e2_closure`.

quantity	value
[0].soil	S03_Sand
[0].method	M02_Surface
[0].crop	vegetable
[0].meanIrrTarget	447
[0].meanIrrSim	0
[0].meanDrainTarget	59.19
[0].meanDrainSim	12.07
[0].resid	5.373
[0].inflow	954.3
[0].relres	0.00563
[0].Esim	4.676
[0].Etarget	5.381
[0].Tsim	29.88
[0].Ttarget	125
[0].kt	1
[1].soil	S10_SiltyClayLoam
[1].method	M04_Drip10-20cm
[1].crop	legume
[1].meanIrrTarget	48.81
[1].meanIrrSim	0
[1].meanDrainTarget	127.3
[1].meanDrainSim	196.8
[1].resid	8.853
[1].inflow	9428
[1].relres	0.0009389
[1].Esim	69.42
[1].Etarget	78.57
[1].Tsim	202.5

quantity	value
[1].Ttarget	252.7
[1].kt	1
[2].soil	S05_SandyLoam
[2].method	M06_Drip20-30cm
[2].crop	vegetable
[2].meanIrrTarget	135.2
[2].meanIrrSim	0
[2].meanDrainTarget	245
[2].meanDrainSim	197.4
[2].resid	7.123
[2].inflow	6486
[2].relres	0.001098
[2].Esim	24.49
[2].Etarget	27.05
[2].Tsim	100.4
[2].Ttarget	176.6
[2].kt	1
[3].soil	S03_Sand
[3].method	M10_Drip50-60cm
[3].crop	vegetable
[3].meanIrrTarget	262.9
[3].meanIrrSim	0
[3].meanDrainTarget	67.23
[3].meanDrainSim	10.26
[3].resid	1.062
[3].inflow	985.7
[3].relres	0.001078
[3].Esim	6.975
[3].Etarget	9.251
[3].Tsim	31.21
[3].Ttarget	156.1
[3].kt	1
[4].soil	S01_CoarseLoam
[4].method	M06_Drip20-30cm
[4].crop	vegetable
[4].meanIrrTarget	197.6
[4].meanIrrSim	0
[4].meanDrainTarget	68.64
[4].meanDrainSim	84.22
[4].resid	8.305
[4].inflow	5360
[4].relres	0.001549
[4].Esim	82.59
[4].Etarget	107.7
[4].Tsim	98.73
[4].Ttarget	281.3
[4].kt	1
[5].soil	S02_CoarseSand

quantity	value
[5].method	M02_Surface
[5].crop	legume
[5].meanIrrTarget	262.6
[5].meanIrrSim	206.1
[5].meanDrainTarget	78.92
[5].meanDrainSim	45.57
[5].resid	2.823
[5].inflow	5097
[5].relres	0.0005539
[5].Esim	9.416
[5].Etarget	13.19
[5].Tsim	199.3
[5].Ttarget	190.5
[5].kt	2.45
[6].soil	S09_ClayLoam
[6].method	M06_Drip20-30cm
[6].crop	vegetable
[6].meanIrrTarget	156.8
[6].meanIrrSim	0
[6].meanDrainTarget	150.8
[6].meanDrainSim	172.8
[6].resid	7.893
[6].inflow	6577
[6].relres	0.0012
[6].Esim	43.68
[6].Etarget	49.53
[6].Tsim	109.7
[6].Ttarget	268.1
[6].kt	1
[7].soil	S07_SiltLoam
[7].method	M10_Drip50-60cm
[7].crop	cereal
[7].meanIrrTarget	325.8
[7].meanIrrSim	327.5
[7].meanDrainTarget	59.89
[7].meanDrainSim	191
[7].resid	7.322
[7].inflow	7526
[7].relres	0.000973
[7].Esim	16.34
[7].Etarget	24.11
[7].Tsim	169.2
[7].Ttarget	86.34
[7].kt	2.681
[8].soil	S05_SandyLoam
[8].method	M02_Surface
[8].crop	vegetable
[8].meanIrrTarget	252.9

quantity	value
[8].meanIrrSim	3.793
[8].meanDrainTarget	44.3
[8].meanDrainSim	0.0631
[8].resid	3.09
[8].inflow	1053
[8].relres	0.002935
[8].Esim	25.18
[8].Etarget	39.34
[8].Tsim	28.81
[8].Ttarget	71.66
[8].kt	1
[9].soil	S09_ClayLoam
[9].method	M06_Drip20-30cm
[9].crop	cereal
[9].meanIrrTarget	188.4
[9].meanIrrSim	182.3
[9].meanDrainTarget	23.79
[9].meanDrainSim	0.08001
[9].resid	2.201
[9].inflow	4761
[9].relres	0.0004624
[9].Esim	28.69
[9].Etarget	83.31
[9].Tsim	208.9
[9].Ttarget	105.8
[9].kt	3.837
[10].soil	S02_CoarseSand
[10].method	M09_Drip40-50cm
[10].crop	legume
[10].meanIrrTarget	258.5
[10].meanIrrSim	9.694
[10].meanDrainTarget	58.34
[10].meanDrainSim	2.344
[10].resid	3.93
[10].inflow	1678
[10].relres	0.002342
[10].Esim	34.03
[10].Etarget	104.1
[10].Tsim	48.23
[10].Ttarget	174.5
[10].kt	6.15
[11].soil	S01_CoarseLoam
[11].method	M10_Drip50-60cm
[11].crop	legume
[11].meanIrrTarget	253.9
[11].meanIrrSim	243.8
[11].meanDrainTarget	57.29
[11].meanDrainSim	138.9

quantity	value
[11].resid	4.042
[11].inflow	7026
[11].relres	0.0005752
[11].Esim	57.37
[11].Etarget	92.84
[11].Tsim	155.4
[11].Ttarget	210.5
[11].kt	0.918
[12].soil	S04_LoamySand
[12].method	M04_Drip10-20cm
[12].crop	legume
[12].meanIrrTarget	448.8
[12].meanIrrSim	432.2
[12].meanDrainTarget	47.68
[12].meanDrainSim	0
[12].resid	4.981
[12].inflow	9519
[12].relres	0.0005233
[12].Esim	6.612
[12].Etarget	7.687
[12].Tsim	469.3
[12].Ttarget	301.2
[12].kt	2.074
[13].soil	S06_Loam
[13].method	M01_Overhead
[13].crop	vegetable
[13].meanIrrTarget	921.7
[13].meanIrrSim	906.5
[13].meanDrainTarget	149.9
[13].meanDrainSim	130.2
[13].resid	5.387
[13].inflow	1.937e+04
[13].relres	0.0002781
[13].Esim	24.66
[13].Etarget	26.94
[13].Tsim	811.3
[13].Ttarget	113.5
[13].kt	7.364
[14].soil	S03_Sand
[14].method	M06_Drip20-30cm
[14].crop	legume
[14].meanIrrTarget	353.8
[14].meanIrrSim	390.2
[14].meanDrainTarget	56.93
[14].meanDrainSim	357.7
[14].resid	5.373
[14].inflow	8759

quantity	value
[14].relres	0.0006134

p3_e2_grid.json

Source: `Manuscripts/results/p3_e2_grid.json`; 1300 values, first 220 shown. Workbook sheet `R_p3_e2_grid`.

quantity	value
[0].soil	S03_Sand
[0].trig	-100
[0].method	M02_Surface
[0].irrT	284.5
[0].irrS	277.4
[0].drT	77.01
[0].drS	130.6
[0].ET	58.22
[0].ES	36.41
[0].TT	207.6
[0].TS	170
[0].kt	1
[0].relres	0.0002886
[1].soil	S03_Sand
[1].trig	-100
[1].method	M01_Overhead
[1].irrT	284.5
[1].irrS	277.4
[1].drT	77.01
[1].drS	130.6
[1].ET	58.22
[1].ES	36.41
[1].TT	207.6
[1].TS	170
[1].kt	1
[1].relres	0.0002886
[2].soil	S03_Sand
[2].trig	-200
[2].method	M02_Surface
[2].irrT	278.2
[2].irrS	285.1
[2].drT	76.52
[2].drS	27.95
[2].ET	58.2
[2].ES	34.19
[2].TT	202.3
[2].TS	289.2

quantity	value
[2].kt	1.843
[2].relres	0.0003502
[3].soil	S03_Sand
[3].trig	-200
[3].method	M01_Overhead
[3].irrT	278.2
[3].irrS	285.1
[3].drT	76.52
[3].drS	27.95
[3].ET	58.2
[3].ES	34.19
[3].TT	202.3
[3].TS	289.2
[3].kt	1.843
[3].relres	0.0003502
[4].soil	S03_Sand
[4].trig	-300
[4].method	M02_Surface
[4].irrT	278.2
[4].irrS	285.1
[4].drT	76.52
[4].drS	5.15
[4].ET	58.2
[4].ES	32.31
[4].TT	202.3
[4].TS	314.8
[4].kt	2.045
[4].relres	0.0003502
[5].soil	S03_Sand
[5].trig	-300
[5].method	M01_Overhead
[5].irrT	278.2
[5].irrS	285.1
[5].drT	76.52
[5].drS	5.15
[5].ET	58.2
[5].ES	32.31
[5].TT	202.3
[5].TS	314.8
[5].kt	2.045
[5].relres	0.0003502
[6].soil	S03_Sand
[6].trig	-500
[6].method	M02_Surface
[6].irrT	278.2
[6].irrS	0
[6].drT	76.52
[6].drS	2.198

quantity	value
[6].ET	58.2
[6].ES	20.58
[6].TT	202.3
[6].TS	44.29
[6].kt	1
[6].relres	0.001843
[7].soil	S03_Sand
[7].trig	-500
[7].method	M01_Overhead
[7].irrT	278.2
[7].irrS	0
[7].drT	76.52
[7].drS	2.198
[7].ET	58.2
[7].ES	20.58
[7].TT	202.3
[7].TS	44.29
[7].kt	1
[7].relres	0.001843
[8].soil	S03_Sand
[8].trig	-1000
[8].method	M02_Surface
[8].irrT	278.2
[8].irrS	0
[8].drT	76.52
[8].drS	2.198
[8].ET	58.2
[8].ES	20.58
[8].TT	202.3
[8].TS	44.29
[8].kt	1
[8].relres	0.001843
[9].soil	S03_Sand
[9].trig	-1000
[9].method	M01_Overhead
[9].irrT	278.2
[9].irrS	0
[9].drT	76.52
[9].drS	2.198
[9].ET	58.2
[9].ES	20.58
[9].TT	202.3
[9].TS	44.29
[9].kt	1
[9].relres	0.001843
[10].soil	S04_LoamySand
[10].trig	-100
[10].method	M02_Surface

quantity	value
[10].irrT	415.9
[10].irrS	411.2
[10].drT	75.78
[10].drS	232.6
[10].ET	55.73
[10].ES	42.95
[10].TT	271.6
[10].TS	237.7
[10].kt	1
[10].relres	0.0005469
[11].soil	S04_LoamySand
[11].trig	-100
[11].method	M01_Overhead
[11].irrT	415.9
[11].irrS	411.2
[11].drT	75.78
[11].drS	232.6
[11].ET	55.73
[11].ES	42.95
[11].TT	271.6
[11].TS	237.7
[11].kt	1
[11].relres	0.0005469
[12].soil	S04_LoamySand
[12].trig	-200
[12].method	M02_Surface
[12].irrT	309.8
[12].irrS	308.3
[12].drT	70.98
[12].drS	3.792
[12].ET	72.1
[12].ES	42.24
[12].TT	224.6
[12].TS	327.8
[12].kt	1.872
[12].relres	0.0003135
[13].soil	S04_LoamySand
[13].trig	-200
[13].method	M01_Overhead
[13].irrT	309.8
[13].irrS	308.3
[13].drT	70.98
[13].drS	3.792
[13].ET	72.1
[13].ES	42.24
[13].TT	224.6
[13].TS	327.8
[13].kt	1.872

quantity	value
[13].relres	0.0003135
[14].soil	S04_LoamySand
[14].trig	-300
[14].method	M02_Surface
[14].irrT	293
[14].irrS	291.6
[14].drT	70.14
[14].drS	0
[14].ET	73.95
[14].ES	41.58
[14].TT	215
[14].TS	319.3
[14].kt	1.93
[14].relres	0.0004296
[15].soil	S04_LoamySand
[15].trig	-300
[15].method	M01_Overhead
[15].irrT	293
[15].irrS	291.6
[15].drT	70.14
[15].drS	0
[15].ET	73.95
[15].ES	41.58
[15].TT	215
[15].TS	319.3
[15].kt	1.93
[15].relres	0.0004296
[16].soil	S04_LoamySand
[16].trig	-500
[16].method	M02_Surface
[16].irrT	293
[16].irrS	337
[16].drT	70.14
[16].drS	0
[16].ET	73.95
[16].ES	31.94
[16].TT	215
[16].TS	374.5
[16].kt	3.837

p3_bench.json

Source: `Manuscripts/results/p3_bench.json`; 4 values. Workbook sheet `R_p3_bench`.

quantity	value
engine1_model_ms	0.0212
sprinkler_ms	1.664
drip_ms	0.251
engine2_s	0.1434

p3_surface.json

Source: `Manuscripts/results/p3_surface.json`; 58 values. Workbook sheet `R_p3_surface`.

quantity	value
first.Treq	378.7
first.rows[0].Q	0.1
first.rows[0].tL	81.65
first.rows[0].r	0.4827
first.rows[0].Ea	0.6516
first.rows[1].Q	0.075
first.rows[1].tL	198.7
first.rows[1].r	0.3852
first.rows[1].Ea	0.6928
first.rows[2].Q	0.0625
first.rows[2].tL	442.3
first.rows[2].r	0.3195
first.rows[2].Ea	0.5846
first.rows[3].Q	0.05
first.rows[3].tL	2091
first.rows[3].r	0.2281
first.rows[3].Ea	0.243
first.rows[4].Q	0.0375
first.rows[4].tL	nan
first.rows[4].r	0.08246
first.rows[4].Ea	0
first.rows[5].Q	0.025
first.rows[5].tL	nan
first.rows[5].r	0.5
first.rows[5].Ea	0
first.rows[6].Q	0.0125
first.rows[6].tL	nan
first.rows[6].r	0.5
first.rows[6].Ea	0
later.Treq	868.5
later.rows[0].Q	0.1
later.rows[0].tL	26.16
later.rows[0].r	0.7569
later.rows[0].Ea	0.3353
later.rows[1].Q	0.075
later.rows[1].tL	36.05

quantity	value
later.rows[1].r	0.6942
later.rows[1].Ea	0.4422
later.rows[2].Q	0.0625
later.rows[2].tL	46.35
later.rows[2].r	0.6459
later.rows[2].Ea	0.5247
later.rows[3].Q	0.05
later.rows[3].tL	68.23
later.rows[3].r	0.5749
later.rows[3].Ea	0.6405
later.rows[4].Q	0.0375
later.rows[4].tL	143.6
later.rows[4].r	0.4532
later.rows[4].Ea	0.7904
later.rows[5].Q	0.025
later.rows[5].tL	nan
later.rows[5].r	0.1755
later.rows[5].Ea	0
later.rows[6].Q	0.0125
later.rows[6].tL	nan
later.rows[6].r	0.5
later.rows[6].Ea	0

p3_surface_shipped.json

Source: `Manuscripts/results/p3_surface_shipped.json`; 104 values. Workbook sheet `R_p3_surface_shipped`.

quantity	value
[0].irrigation	first
[0].Q	0.1
[0].A0	0.007773
[0].r	0.4827
[0].tL_book	80.2
[0].tL_reference	81.65
[0].tL_shipped	81.66
[0].tL_shipped_rAdv_r	81.66
[0].tL_v100	78.81
[0].err_reference_pct	1.814
[0].err_shipped_pct	1.815
[0].err_shipped_rAdv_r_pct	1.815
[0].err_v100_pct	-1.729
[1].irrigation	first
[1].Q	0.075
[1].A0	0.006298

quantity	value
[1].r	0.3852
[1].tL_book	198.6
[1].tL_reference	198.7
[1].tL_shipped	198.7
[1].tL_shipped_rAdv_r	198.7
[1].tL_v100	169.5
[1].err_reference_pct	0.03707
[1].err_shipped_pct	0.03792
[1].err_shipped_rAdv_r_pct	0.03782
[1].err_v100_pct	-14.67
[2].irrigation	first
[2].Q	0.0625
[2].A0	0.005511
[2].r	0.3195
[2].tL_book	441.6
[2].tL_reference	442.3
[2].tL_shipped	442.3
[2].tL_shipped_rAdv_r	442.3
[2].tL_v100	313.7
[2].err_reference_pct	0.1522
[2].err_shipped_pct	0.153
[2].err_shipped_rAdv_r_pct	0.1528
[2].err_v100_pct	-28.96
[3].irrigation	later
[3].Q	0.1
[3].A0	0.007773
[3].r	0.7569
[3].tL_book	26.2
[3].tL_reference	26.16
[3].tL_shipped	26.16
[3].tL_shipped_rAdv_r	26.16
[3].tL_v100	27.07
[3].err_reference_pct	-0.1565
[3].err_shipped_pct	-0.1566
[3].err_shipped_rAdv_r_pct	-0.1566
[3].err_v100_pct	3.338
[4].irrigation	later
[4].Q	0.075
[4].A0	0.006298
[4].r	0.6942
[4].tL_book	36
[4].tL_reference	36.05
[4].tL_shipped	36.05
[4].tL_shipped_rAdv_r	36.05
[4].tL_v100	37.42
[4].err_reference_pct	0.1366
[4].err_shipped_pct	0.1365
[4].err_shipped_rAdv_r_pct	0.1365

quantity	value
[4].err_v100_pct	3.957
[5].irrigation	later
[5].Q	0.0625
[5].A0	0.005511
[5].r	0.6459
[5].tL_book	46.3
[5].tL_reference	46.35
[5].tL_shipped	46.35
[5].tL_shipped_rAdv_r	46.35
[5].tL_v100	47.95
[5].err_reference_pct	0.1122
[5].err_shipped_pct	0.1119
[5].err_shipped_rAdv_r_pct	0.1119
[5].err_v100_pct	3.574
[6].irrigation	later
[6].Q	0.05
[6].A0	0.004681
[6].r	0.5749
[6].tL_book	68.2
[6].tL_reference	68.23
[6].tL_shipped	68.23
[6].tL_shipped_rAdv_r	68.23
[6].tL_v100	69.16
[6].err_reference_pct	0.04479
[6].err_shipped_pct	0.04405
[6].err_shipped_rAdv_r_pct	0.04414
[6].err_v100_pct	1.404
[7].irrigation	later
[7].Q	0.0375
[7].A0	0.003793
[7].r	0.4532
[7].tL_book	143.5
[7].tL_reference	143.6
[7].tL_shipped	143.6
[7].tL_shipped_rAdv_r	143.6
[7].tL_v100	129.4
[7].err_reference_pct	0.0976
[7].err_shipped_pct	0.09911
[7].err_shipped_rAdv_r_pct	0.0988
[7].err_v100_pct	-9.814

p3_border.json

Source: `Manuscripts/results/p3_border.json`; 10 values. Workbook sheet `R_p3_border`.

quantity	value
book.Treq_first	146.4
book.Treq_later	162.8
book.Qmax	1.88
book.tL_first_Q014_L360	158
book.A0	0.0303
computed.Treq_first	146.4
computed.Treq_later	162.2
computed.Qmax	1.883
computed.tL_first_Q014_L360	158.7
computed.A0	0.03034

p3_sprinkler.json

Source: `Manuscripts/results/p3_sprinkler.json`; 123 values. Workbook sheet `R_p3_sprinkler`.

quantity	value
0.675.umax	8
0.675.qmax	1.44
0.675.tset	16
0.675.sets	1
0.675.N	15
0.675.Nm	12
0.675.Tint	10
0.675.opts[0].n	1
0.675.opts[0].Nd	12
0.675.opts[0].Qin	10.12
0.675.opts[0].okT	False
0.675.opts[0].okQ	True
0.675.opts[1].n	2
0.675.opts[1].Nd	6
0.675.opts[1].Qin	20.25
0.675.opts[1].okT	True
0.675.opts[1].okQ	True
0.675.opts[2].n	3
0.675.opts[2].Nd	4
0.675.opts[2].Qin	30.37
0.675.opts[2].okT	True
0.675.opts[2].okQ	False
0.675.opts[3].n	4
0.675.opts[3].Nd	3
0.675.opts[3].Qin	40.5
0.675.opts[3].okT	True
0.675.opts[3].okQ	False
0.675.pick	2
0.675.Rw_ok	1
0.675.chk[0]	Spacing along the lateral S ≤ k·Rw, True

quantity	value
0.675.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
0.675.chk[2]	Wind below 16 km/h, True
0.675.chk[3]	Application rate r ≤ maximum allowed, True
0.675.chk[4]	Rate below the soil’s saturated conductivity, True
0.675.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
0.675.chk[6]	Lateral head loss ≤ 20 /% of hs, True
0.675.chk[7]	Mainline gradient J < 0.05, True
0.675.chk[8]	Whole field done within the interval T, True
0.675.chk[9]	Flow within the farm supply, True
1.35.umax	8
1.35.qmax	1.44
1.35.tset	8
1.35.sets	2
1.35.N	15
1.35.Nm	12
1.35.Tint	10
1.35.opts[0].n	1
1.35.opts[0].Nd	6
1.35.opts[0].Qin	20.25
1.35.opts[0].okT	True
1.35.opts[0].okQ	True
1.35.opts[1].n	2
1.35.opts[1].Nd	3
1.35.opts[1].Qin	40.5
1.35.opts[1].okT	True
1.35.opts[1].okQ	False
1.35.opts[2].n	3
1.35.opts[2].Nd	2
1.35.opts[2].Qin	60.75
1.35.opts[2].okT	True
1.35.opts[2].okQ	False
1.35.opts[3].n	4
1.35.opts[3].Nd	2
1.35.opts[3].Qin	81
1.35.opts[3].okT	True
1.35.opts[3].okQ	False
1.35.pick	1
1.35.Rw_ok	1
1.35.chk[0]	Spacing along the lateral S ≤ k·Rw, True
1.35.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
1.35.chk[2]	Wind below 16 km/h, True
1.35.chk[3]	Application rate r ≤ maximum allowed, True
1.35.chk[4]	Rate below the soil’s saturated conductivity, True
1.35.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
1.35.chk[6]	Lateral head loss ≤ 20 /% of hs, True
1.35.chk[7]	Mainline gradient J < 0.05, True
1.35.chk[8]	Whole field done within the interval T, True
1.35.chk[9]	Flow within the farm supply, True

quantity	value
2.03.umax	8
2.03.qmax	1.44
2.03.tset	5.32
2.03.sets	3
2.03.N	15
2.03.Nm	12
2.03.Tint	10
2.03.opts[0].n	1
2.03.opts[0].Nd	4
2.03.opts[0].Qin	30.45
2.03.opts[0].okT	True
2.03.opts[0].okQ	False
2.03.opts[1].n	2
2.03.opts[1].Nd	2
2.03.opts[1].Qin	60.9
2.03.opts[1].okT	True
2.03.opts[1].okQ	False
2.03.opts[2].n	3
2.03.opts[2].Nd	2
2.03.opts[2].Qin	91.35
2.03.opts[2].okT	True
2.03.opts[2].okQ	False
2.03.opts[3].n	4
2.03.opts[3].Nd	1
2.03.opts[3].Qin	121.8
2.03.opts[3].okT	True
2.03.opts[3].okQ	False
2.03.pick	1
2.03.Rw_ok	1
2.03.chk[0]	Spacing along the lateral S ≤ k·Rw, True
2.03.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
2.03.chk[2]	Wind below 16 km/h, True
2.03.chk[3]	Application rate r ≤ maximum allowed, False
2.03.chk[4]	Rate below the soil’s saturated conductivity, True
2.03.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
2.03.chk[6]	Lateral head loss ≤ 20 /% of hs, True
2.03.chk[7]	Mainline gradient J < 0.05, True
2.03.chk[8]	Whole field done within the interval T, True
2.03.chk[9]	Flow within the farm supply, False
nozzle.q	22.38
nozzle.R	14.49
nozzle.h	24
nozzle_check.q	22.38
nozzle_check.R	14.49
nozzle_check.h	24

p3_drip_swafl.json

Source: `Manuscripts/results/p3_drip_swafl.json`; 960 values, first 220 shown. Workbook sheet `R_p3_drip_swafl`.

quantity	value
[0].kind	online
[0].q	3.773
[0].Se	1.296
[0].LL	138
[0].H0	19.12
[0].D	13
[0].x	0.5
[0].Hmin	0
[0].cv	0.07
[0].ne	1
[0].hf	3.44
[0].Hend	15.68
[0].Eu	0.8865
[0].qa	3.512
[0].qmin	3.417
[0].N	106
[1].kind	inline
[1].q	2.627
[1].Se	0.6332
[1].LL	48.5
[1].H0	8.825
[1].D	25
[1].x	0.5
[1].Hmin	0
[1].cv	0.04
[1].ne	1
[1].hf	0.01682
[1].Hend	8.809
[1].Eu	0.949
[1].qa	2.625
[1].qmin	2.624
[1].N	77
[2].kind	tape
[2].q	1.384
[2].Se	0.4125
[2].LL	112.3
[2].H0	14.57
[2].D	20
[2].x	0.6
[2].Hmin	0
[2].cv	0.08
[2].ne	2

quantity	value
[2].hf	0.3499
[2].Hend	14.22
[2].Eu	0.9246
[2].qa	1.369
[2].qmin	1.364
[2].N	272
[3].kind	online
[3].q	5.066
[3].Se	1.136
[3].LL	106.9
[3].H0	13.69
[3].D	25
[3].x	0.5
[3].Hmin	0
[3].cv	0.07
[3].ne	1
[3].hf	0.1776
[3].Hend	13.51
[3].Eu	0.9095
[3].qa	5.042
[3].qmin	5.033
[3].N	94
[4].kind	inline
[4].q	2.035
[4].Se	0.7778
[4].LL	54.71
[4].H0	16.99
[4].D	25
[4].x	0.5
[4].Hmin	0
[4].cv	0.04
[4].ne	1
[4].hf	0.009837
[4].Hend	16.98
[4].Eu	0.9491
[4].qa	2.035
[4].qmin	2.034
[4].N	70
[5].kind	tape
[5].q	1.015
[5].Se	0.2715
[5].LL	197.1
[5].H0	9.465
[5].D	16
[5].x	0.6
[5].Hmin	0
[5].cv	0.08
[5].ne	2

quantity	value
[5].hf	3.295
[5].Hend	6.169
[5].Eu	0.8636
[5].qa	0.8437
[5].qmin	0.785
[5].N	726
[6].kind	online
[6].q	4.295
[6].Se	0.3296
[6].LL	58.24
[6].H0	15.46
[6].D	25
[6].x	0.5
[6].Hmin	0
[6].cv	0.07
[6].ne	1
[6].hf	0.2288
[6].Hend	15.23
[6].Eu	0.9093
[6].qa	4.272
[6].qmin	4.263
[6].N	177
[7].kind	inline
[7].q	1.701
[7].Se	0.9267
[7].LL	188.2
[7].H0	11.19
[7].D	20
[7].x	0.5
[7].Hmin	0
[7].cv	0.04
[7].ne	1
[7].hf	0.4948
[7].Hend	10.69
[7].Eu	0.9435
[7].qa	1.673
[7].qmin	1.663
[7].N	203
[8].kind	tape
[8].q	1.012
[8].Se	0.3251
[8].LL	122
[8].H0	16.54
[8].D	16
[8].x	0.6
[8].Hmin	0
[8].cv	0.08
[8].ne	2

quantity	value
[8].hf	1.095
[8].Hend	15.45
[8].Eu	0.9181
[8].qa	0.9815
[8].qmin	0.9709
[8].N	375
[9].kind	online
[9].q	3.58
[9].Se	1.249
[9].LL	140.1
[9].H0	12.55
[9].D	25
[9].x	0.5
[9].Hmin	0
[9].cv	0.07
[9].ne	1
[9].hf	0.1687
[9].Hend	12.38
[9].Eu	0.9095
[9].qa	3.562
[9].qmin	3.556
[9].N	112
[10].kind	inline
[10].q	2.181
[10].Se	0.4138
[10].LL	80.59
[10].H0	19.05
[10].D	20
[10].x	0.5
[10].Hmin	0
[10].cv	0.04
[10].ne	1
[10].hf	0.3194
[10].Hend	18.73
[10].Eu	0.9471
[10].qa	2.168
[10].qmin	2.163
[10].N	195
[11].kind	tape
[11].q	1.096
[11].Se	0.27
[11].LL	109.4
[11].H0	10.14
[11].D	13
[11].x	0.6
[11].Hmin	0
[11].cv	0.08
[11].ne	2

quantity	value
[11].hf	2.779
[11].Hend	7.364
[11].Eu	0.8794
[11].qa	0.9542
[11].qmin	0.9041
[11].N	405
[12].kind	online
[12].q	4.86
[12].Se	1.112
[12].LL	63.15
[12].H0	14.52
[12].D	20
[12].x	0.5
[12].Hmin	0
[12].cv	0.07
[12].ne	1
[12].hf	0.1162
[12].Hend	14.4
[12].Eu	0.9101
[12].qa	4.846
[12].qmin	4.84
[12].N	57
[13].kind	inline
[13].q	2.659
[13].Se	0.6456
[13].LL	105.1
[13].H0	15.66
[13].D	25
[13].x	0.5
[13].Hmin	0
[13].cv	0.04
[13].ne	1
[13].hf	0.1466
[13].Hend	15.51

p3_drip_bands2.json

Source: `Manuscripts/results/p3_drip_bands2.json`; 6 values. Workbook sheet `R_p3_drip_bands2`.

quantity	value
new -1-0.2	244, 5.341e-09, 3.21e-07, 2.644e-08
new 0.2-0.6	80, 1.138e-07, 2.549e-07, 9.205e-08
new 0.6-2	76, 4.764e-16, 0.3092, 0.0797
old -1-0.2	244, 1.025e-08, 0.001344, 4.567e-05
old 0.2-0.6	80, 0.01911, 0.8116, 0.1326

quantity	value
old 0.6-2	76, 0.6731, 0.8912, 0.4438

p3_ws61.csv

Source: `Manuscripts/results/p3_ws61.csv`; 16 rows. Workbook sheet `R_p3_ws61`.

irr	quantity	book	swafl	err_pct
first	Treq	378.7	378.7	0.0007214
first	tL Q=0.1	80.2	81.65	1.814
first	Ea Q=0.1	65.4	65.16	-0.3631
first	tL Q=0.075	198.6	198.7	0.03707
first	Ea Q=0.075	69.3	69.28	-0.03564
first	tL Q=0.0625	441.6	442.3	0.1522
first	Ea Q=0.0625	58.5	58.46	-0.05991
later	Treq	868.5	868.5	0.002155
later	tL Q=0.1	26.2	26.16	-0.1565
later	tL Q=0.075	36	36.05	0.1366
later	Ea Q=0.075	44.2	44.22	0.04527
later	tL Q=0.0625	46.3	46.35	0.1122
later	tL Q=0.05	68.2	68.23	0.04479
later	Ea Q=0.05	64.1	64.05	-0.07599
later	tL Q=0.0375	143.5	143.6	0.0976
later	Ea Q=0.0375	79.1	79.04	-0.07714

fao_out.json

Source: `Manuscripts/results/fao_out.json`; 98 values. Workbook sheet `R_fao_out`.

quantity	value
[0].id	first
[0].treq	388.5
[0].treqRef	388.5
[0].qm	0.02258
[0].qmRef	0.0226
[0].rows[0].q	0.036
[0].rows[0].ref	6.36
[0].rows[0].adv	6.357
[0].rows[0].err	-0.0539
[0].rows[0].adv_v100	6.391
[0].rows[0].err_v100	0.4827
[0].rows[1].q	0.0545
[0].rows[1].ref	3.11
[0].rows[1].adv	3.119

quantity	value
[0].rows[1].err	0.2913
[0].rows[1].adv_v100	3.188
[0].rows[1].err_v100	2.519
[0].rows[2].q	0.072
[0].rows[2].ref	2.14
[0].rows[2].adv	2.135
[0].rows[2].err	-0.2347
[0].rows[2].adv_v100	2.182
[0].rows[2].err_v100	1.961
[0].rows[3].q	0.09
[0].rows[3].ref	1.64
[0].rows[3].adv	1.641
[0].rows[3].err	0.0496
[0].rows[3].adv_v100	1.673
[0].rows[3].err_v100	2.036
[1].id	later
[1].treq	678.9
[1].treqRef	678.9
[1].qm	0.02258
[1].qmRef	0.0226
[1].rows[0].q	0.018
[1].rows[0].ref	15.55
[1].rows[0].adv	15.56
[1].rows[0].err	0.03386
[1].rows[0].adv_v100	15.57
[1].rows[0].err_v100	0.1467
[1].rows[1].q	0.036
[1].rows[1].ref	5.03
[1].rows[1].adv	5.028
[1].rows[1].err	-0.04831
[1].rows[1].adv_v100	5.176
[1].rows[1].err_v100	2.904
[1].rows[2].q	0.0545
[1].rows[2].ref	3.15
[1].rows[2].adv	3.154
[1].rows[2].err	0.1409
[1].rows[2].adv_v100	3.228
[1].rows[2].err_v100	2.465
[2].id	first100
[2].treq	388.5
[2].treqRef	388.5
[2].qm	0.03193
[2].qmRef	0.032
[2].rows[0].q	0.018
[2].rows[0].ref	5.27
[2].rows[0].adv	5.268
[2].rows[0].err	-0.04728
[2].rows[0].adv_v100	5.188

quantity	value
[2].rows[0].err_v100	-1.548
[2].rows[1].q	0.0269
[2].rows[1].ref	2.35
[2].rows[1].adv	2.348
[2].rows[1].err	-0.09153
[2].rows[1].adv_v100	2.371
[2].rows[1].err_v100	0.9027
[2].rows[2].q	0.036
[2].rows[2].ref	1.44
[2].rows[2].adv	1.442
[2].rows[2].err	0.1328
[2].rows[2].adv_v100	1.462
[2].rows[2].err_v100	1.554
[2].rows[3].q	0.045
[2].rows[3].ref	1.03
[2].rows[3].adv	1.035
[2].rows[3].err	0.4668
[2].rows[3].adv_v100	1.05
[2].rows[3].err_v100	1.92
[3].id	later100
[3].treq	678.9
[3].treqRef	678.9
[3].qm	0.03193
[3].qmRef	0.032
[3].rows[0].q	0.009
[3].rows[0].ref	12.89
[3].rows[0].adv	12.89
[3].rows[0].err	0.03702
[3].rows[0].adv_v100	12.61
[3].rows[0].err_v100	-2.198
[3].rows[1].q	0.018
[3].rows[1].ref	3.45
[3].rows[1].adv	3.446
[3].rows[1].err	-0.1056
[3].rows[1].adv_v100	3.52
[3].rows[1].err_v100	2.034

spr_out.json

Source: `Manuscripts/results/spr_out.json`; 432 values, first 220 shown. Workbook sheet `R_spr_out`.

quantity	value
new.nozzle.q	22.38
new.nozzle.R	14.49
new.nozzle.h	24
new.D	28.98

quantity	value
new.cu[0].u	0
new.cu[0].ratio	0.4
new.cu[0].S	11.59
new.cu[0].CU	0.974
new.cu[0].AR	9.992
new.cu[0].Ea	0.9873
new.cu[1].u	0
new.cu[1].ratio	0.5
new.cu[1].S	14.49
new.cu[1].CU	0.9435
new.cu[1].AR	6.392
new.cu[1].Ea	0.9873
new.cu[2].u	0
new.cu[2].ratio	0.6
new.cu[2].S	17.39
new.cu[2].CU	0.9289
new.cu[2].AR	4.445
new.cu[2].Ea	0.9873
new.cu[3].u	0
new.cu[3].ratio	0.65
new.cu[3].S	18.84
new.cu[3].CU	0.8573
new.cu[3].AR	3.785
new.cu[3].Ea	0.9873
new.cu[4].u	0
new.cu[4].ratio	0.7
new.cu[4].S	20.29
new.cu[4].CU	0.7868
new.cu[4].AR	3.261
new.cu[4].Ea	0.9873
new.cu[5].u	0
new.cu[5].ratio	0.8
new.cu[5].S	23.18
new.cu[5].CU	0.6247
new.cu[5].AR	2.5
new.cu[5].Ea	0.9873
new.cu[6].u	0
new.cu[6].ratio	1
new.cu[6].S	28.98
new.cu[6].CU	0.2207
new.cu[6].AR	1.6
new.cu[6].Ea	0.9873
new.cu[7].u	0
new.cu[7].ratio	1.2
new.cu[7].S	34.78
new.cu[7].CU	-0.08095
new.cu[7].AR	1.109
new.cu[7].Ea	0.9873

quantity	value
new.cu[8].u	3
new.cu[8].ratio	0.4
new.cu[8].S	11.59
new.cu[8].CU	0.9744
new.cu[8].AR	9.993
new.cu[8].Ea	0.9677
new.cu[9].u	3
new.cu[9].ratio	0.5
new.cu[9].S	14.49
new.cu[9].CU	0.9416
new.cu[9].AR	6.394
new.cu[9].Ea	0.9677
new.cu[10].u	3
new.cu[10].ratio	0.6
new.cu[10].S	17.39
new.cu[10].CU	0.9281
new.cu[10].AR	4.443
new.cu[10].Ea	0.9677
new.cu[11].u	3
new.cu[11].ratio	0.65
new.cu[11].S	18.84
new.cu[11].CU	0.8581
new.cu[11].AR	3.785
new.cu[11].Ea	0.9677
new.cu[12].u	3
new.cu[12].ratio	0.7
new.cu[12].S	20.29
new.cu[12].CU	0.7877
new.cu[12].AR	3.262
new.cu[12].Ea	0.9677
new.cu[13].u	3
new.cu[13].ratio	0.8
new.cu[13].S	23.18
new.cu[13].CU	0.6262
new.cu[13].AR	2.5
new.cu[13].Ea	0.9677
new.cu[14].u	3
new.cu[14].ratio	1
new.cu[14].S	28.98
new.cu[14].CU	0.2226
new.cu[14].AR	1.6
new.cu[14].Ea	0.9677
new.cu[15].u	3
new.cu[15].ratio	1.2
new.cu[15].S	34.78
new.cu[15].CU	-0.07702
new.cu[15].AR	1.11
new.cu[15].Ea	0.9677

quantity	value
new.cutri[0].ratio	0.5
new.cutri[0].CU	0.9463
new.cutri[1].ratio	0.6
new.cutri[1].CU	0.937
new.cutri[2].ratio	0.7
new.cutri[2].CU	0.9279
new.cutri[3].ratio	0.8
new.cutri[3].CU	0.7518
new.q[0].d	2.4
new.q[0].P	200
new.q[0].q	5.157
new.q[0].qth	5.157
new.q[0].R	9.443
new.q[0].Rkb	9.443
new.q[1].d	2.4
new.q[1].P	300
new.q[1].q	6.316
new.q[1].qth	6.316
new.q[1].R	11.57
new.q[1].Rkb	11.57
new.q[2].d	2.4
new.q[2].P	400
new.q[2].q	7.293
new.q[2].qth	7.293
new.q[2].R	13.35
new.q[2].Rkb	13.35
new.q[3].d	3.2
new.q[3].P	200
new.q[3].q	9.168
new.q[3].qth	9.168
new.q[3].R	10.9
new.q[3].Rkb	10.9
new.q[4].d	3.2
new.q[4].P	300
new.q[4].q	11.23
new.q[4].qth	11.23
new.q[4].R	13.35
new.q[4].Rkb	13.35
new.q[5].d	3.2
new.q[5].P	400
new.q[5].q	12.97
new.q[5].qth	12.97
new.q[5].R	15.42
new.q[5].Rkb	15.42
new.q[6].d	4
new.q[6].P	200
new.q[6].q	14.33
new.q[6].qth	14.33

quantity	value
new.q[6].R	12.19
new.q[6].Rkb	12.19
new.q[7].d	4
new.q[7].P	300
new.q[7].q	17.55
new.q[7].qth	17.55
new.q[7].R	14.93
new.q[7].Rkb	14.93
new.q[8].d	4
new.q[8].P	400
new.q[8].q	20.26
new.q[8].qth	20.26
new.q[8].R	17.24
new.q[8].Rkb	17.24
new.q[9].d	4.8
new.q[9].P	200
new.q[9].q	20.63
new.q[9].qth	20.63
new.q[9].R	13.35
new.q[9].Rkb	13.35
new.q[10].d	4.8
new.q[10].P	300
new.q[10].q	25.27
new.q[10].qth	25.27
new.q[10].R	16.36
new.q[10].Rkb	16.36
new.q[11].d	4.8
new.q[11].P	400
new.q[11].q	29.17
new.q[11].qth	29.17
new.q[11].R	18.89
new.q[11].Rkb	18.89
new.q[12].d	6
new.q[12].P	200
new.q[12].q	32.23
new.q[12].qth	32.23
new.q[12].R	14.93
new.q[12].Rkb	14.93
new.q[13].d	6
new.q[13].P	300
new.q[13].q	39.48
new.q[13].qth	39.48
new.q[13].R	18.29
new.q[13].Rkb	18.29
new.q[14].d	6
new.q[14].P	400
new.q[14].q	45.58
new.q[14].qth	45.58

quantity	value
new.q[14].R	21.12
new.q[14].Rkb	21.12
new.q[15].d	8
new.q[15].P	200
new.q[15].q	57.3
new.q[15].qth	57.3
new.q[15].R	17.24
new.q[15].Rkb	17.24
new.q[16].d	8
new.q[16].P	300
new.q[16].q	70.18
new.q[16].qth	70.18
new.q[16].R	21.12
new.q[16].Rkb	21.12
new.q[17].d	8
new.q[17].P	400
new.q[17].q	81.04
new.q[17].qth	81.04
new.q[17].R	24.38
new.q[17].Rkb	24.38
old.nozzle.q	22.38
old.nozzle.R	12.34
old.nozzle.h	24
old.D	24.69

drip_out.json

Source: `Manuscripts/results/drip_out.json`; 14470 values, first 220 shown. Workbook sheet `R_drip_out`.

quantity	value
new.rows[0].kind	online
new.rows[0].q	3.265
new.rows[0].Se	0.421
new.rows[0].LL	124.9
new.rows[0].H0	7.351
new.rows[0].D	25
new.rows[0].x	0.5
new.rows[0].Hmin	0
new.rows[0].cv	0.07
new.rows[0].ne	1
new.rows[0].N	297
new.rows[0].hf	0.7154
new.rows[0].Hend	6.635
new.rows[0].Eu	0.8987

quantity	value
new.rows[0].qa	3.145
new.rows[0].qmin	3.102
new.rows[0].conv	True
new.rows[0].iters	9
new.rows[0].hfFrac	0.09732
new.rows[0].warn	False
new.rows[1].kind	inline
new.rows[1].q	2.724
new.rows[1].Se	0.2681
new.rows[1].LL	223.8
new.rows[1].H0	7.866
new.rows[1].D	13
new.rows[1].x	0.5
new.rows[1].Hmin	0
new.rows[1].cv	0.04
new.rows[1].ne	1
new.rows[1].N	835
new.rows[1].hf	7.366
new.rows[1].Hend	0.5
new.rows[1].Eu	0.7249
new.rows[1].qa	0.8993
new.rows[1].qmin	0.6868
new.rows[1].conv	True
new.rows[1].iters	25
new.rows[1].hfFrac	0.9364
new.rows[1].warn	True
new.rows[2].kind	tape
new.rows[2].q	1.309
new.rows[2].Se	0.4164
new.rows[2].LL	127
new.rows[2].H0	12.39
new.rows[2].D	20
new.rows[2].x	0.6
new.rows[2].Hmin	0
new.rows[2].cv	0.08
new.rows[2].ne	2
new.rows[2].N	305
new.rows[2].hf	0.4366
new.rows[2].Hend	11.96
new.rows[2].Eu	0.9229
new.rows[2].qa	1.288
new.rows[2].qmin	1.281
new.rows[2].conv	True
new.rows[2].iters	5
new.rows[2].hfFrac	0.03523
new.rows[2].warn	False
new.rows[3].kind	pc
new.rows[3].q	2.355

quantity	value
new.rows[3].Se	0.5139
new.rows[3].LL	159.3
new.rows[3].H0	18.03
new.rows[3].D	16
new.rows[3].x	0.05
new.rows[3].Hmin	8
new.rows[3].cv	0.03
new.rows[3].ne	1
new.rows[3].N	310
new.rows[3].hf	5.049
new.rows[3].Hend	12.98
new.rows[3].Eu	0.9591
new.rows[3].qa	2.336
new.rows[3].qmin	2.329
new.rows[3].conv	True
new.rows[3].iters	4
new.rows[3].hfFrac	0.28
new.rows[3].warn	True
new.rows[4].kind	online
new.rows[4].q	2.857
new.rows[4].Se	0.7863
new.rows[4].LL	238.7
new.rows[4].H0	6.648
new.rows[4].D	13
new.rows[4].x	0.5
new.rows[4].Hmin	0
new.rows[4].cv	0.07
new.rows[4].ne	1
new.rows[4].N	304
new.rows[4].hf	6.148
new.rows[4].Hend	0.5
new.rows[4].Eu	0.6143
new.rows[4].qa	1.393
new.rows[4].qmin	0.9395
new.rows[4].conv	False
new.rows[4].iters	500
new.rows[4].hfFrac	0.9248
new.rows[4].warn	True
new.rows[5].kind	inline
new.rows[5].q	1.426
new.rows[5].Se	0.6609
new.rows[5].LL	43.52
new.rows[5].H0	12.51
new.rows[5].D	32
new.rows[5].x	0.5
new.rows[5].Hmin	0
new.rows[5].cv	0.04
new.rows[5].ne	1

quantity	value
new.rows[5].N	66
new.rows[5].hf	0.0011
new.rows[5].Hend	12.51
new.rows[5].Eu	0.9492
new.rows[5].qa	1.426
new.rows[5].qmin	1.426
new.rows[5].conv	True
new.rows[5].iters	4
new.rows[5].hfFrac	8.798e-05
new.rows[5].warn	False
new.rows[6].kind	tape
new.rows[6].q	1.326
new.rows[6].Se	0.7828
new.rows[6].LL	291.2
new.rows[6].H0	6.849
new.rows[6].D	16
new.rows[6].x	0.6
new.rows[6].Hmin	0
new.rows[6].cv	0.08
new.rows[6].ne	2
new.rows[6].N	372
new.rows[6].hf	2.78
new.rows[6].Hend	4.069
new.rows[6].Eu	0.8466
new.rows[6].qa	1.064
new.rows[6].qmin	0.9702
new.rows[6].conv	True
new.rows[6].iters	15
new.rows[6].hfFrac	0.406
new.rows[6].warn	True
new.rows[7].kind	pc
new.rows[7].q	2.596
new.rows[7].Se	0.2255
new.rows[7].LL	87.88
new.rows[7].H0	21.47
new.rows[7].D	13
new.rows[7].x	0.05
new.rows[7].Hmin	8
new.rows[7].cv	0.03
new.rows[7].ne	1
new.rows[7].N	390
new.rows[7].hf	13.8
new.rows[7].Hend	7.669
new.rows[7].Eu	0.9533
new.rows[7].qa	2.554
new.rows[7].qmin	2.531
new.rows[7].conv	True
new.rows[7].iters	22

quantity	value
new.rows[7].hfFrac	0.6428
new.rows[7].warn	True
new.rows[8].kind	online
new.rows[8].q	3.418
new.rows[8].Se	0.897
new.rows[8].LL	232.7
new.rows[8].H0	22.88
new.rows[8].D	25
new.rows[8].x	0.5
new.rows[8].Hmin	0
new.rows[8].cv	0.07
new.rows[8].ne	1
new.rows[8].N	259
new.rows[8].hf	1.169
new.rows[8].Hend	21.71
new.rows[8].Eu	0.9048
new.rows[8].qa	3.353
new.rows[8].qmin	3.329
new.rows[8].conv	True
new.rows[8].iters	5
new.rows[8].hfFrac	0.05109
new.rows[8].warn	False
new.rows[9].kind	inline
new.rows[9].q	1.478
new.rows[9].Se	0.5697
new.rows[9].LL	190.2
new.rows[9].H0	16.27
new.rows[9].D	16
new.rows[9].x	0.5
new.rows[9].Hmin	0
new.rows[9].cv	0.04
new.rows[9].ne	1
new.rows[9].N	334
new.rows[9].hf	2.59
new.rows[9].Hend	13.68
new.rows[9].Eu	0.9271
new.rows[9].qa	1.388
new.rows[9].qmin	1.356
new.rows[9].conv	True
new.rows[9].iters	12
new.rows[9].hfFrac	0.1591
new.rows[9].warn	False
new.rows[10].kind	tape
new.rows[10].q	0.8285
new.rows[10].Se	0.694
new.rows[10].LL	153
new.rows[10].H0	6.85
new.rows[10].D	25

quantity	value
new.rows[10].x	0.6
new.rows[10].Hmin	0
new.rows[10].cv	0.08
new.rows[10].ne	2
new.rows[10].N	220
new.rows[10].hf	0.04271
new.rows[10].Hend	6.807
new.rows[10].Eu	0.9272
new.rows[10].qa	0.8262
new.rows[10].qmin	0.8254
new.rows[10].conv	True
new.rows[10].iters	4
new.rows[10].hfFrac	0.006235
new.rows[10].warn	False

e12.json

Source: `Manuscripts/results/e12.json`; 12000 values, first 220 shown. Workbook sheet `R_e12`.

quantity	value
[0].trig	-100
[0].z	100
[0].P	0.5
[0].ET_mm	674.7
[0].irrigation_mm	438.2
[0].drainage_1m_mm	3.799
[0].yield_Mg_ha	4.935
[0].transpiration_mm	244.4
[0].soil_evap_mm	97.75
[0].N_leached_kg_ha	43.62
[0].treatment	z100cm_h100_P0.5
[0].soil	S01_CoarseLoam
[1].trig	-100
[1].z	100
[1].P	0.6
[1].ET_mm	712.6
[1].irrigation_mm	436.6
[1].drainage_1m_mm	8.117
[1].yield_Mg_ha	6.598
[1].transpiration_mm	263.4
[1].soil_evap_mm	108.6
[1].N_leached_kg_ha	10.62
[1].treatment	z100cm_h100_P0.6
[1].soil	S01_CoarseLoam
[2].trig	-100
[2].z	100

quantity	value
[2].P	0.7
[2].ET_mm	715.4
[2].irrigation_mm	408.6
[2].drainage_1m_mm	22.13
[2].yield_Mg_ha	6.625
[2].transpiration_mm	256.6
[2].soil_evap_mm	116.3
[2].N_leached_kg_ha	14.91
[2].treatment	z100cm_h100_P0.7
[2].soil	S01_CoarseLoam
[3].trig	-100
[3].z	100
[3].P	0.8
[3].ET_mm	704.6
[3].irrigation_mm	358.6
[3].drainage_1m_mm	31.85
[3].yield_Mg_ha	6.355
[3].transpiration_mm	242.8
[3].soil_evap_mm	127.5
[3].N_leached_kg_ha	23.82
[3].treatment	z100cm_h100_P0.8
[3].soil	S01_CoarseLoam
[4].trig	-100
[4].z	100
[4].P	0.9
[4].ET_mm	705.5
[4].irrigation_mm	324.7
[4].drainage_1m_mm	42.4
[4].yield_Mg_ha	6.397
[4].transpiration_mm	240.8
[4].soil_evap_mm	134.5
[4].N_leached_kg_ha	25.07
[4].treatment	z100cm_h100_P0.9
[4].soil	S01_CoarseLoam
[5].trig	-100
[5].z	100
[5].P	1
[5].ET_mm	699.6
[5].irrigation_mm	289.1
[5].drainage_1m_mm	59.81
[5].yield_Mg_ha	6.263
[5].transpiration_mm	232.8
[5].soil_evap_mm	140.7
[5].N_leached_kg_ha	30.91
[5].treatment	z100cm_h100_P1
[5].soil	S01_CoarseLoam
[6].trig	-100
[6].z	100

quantity	value
[6].P	1.1
[6].ET_mm	696.8
[6].irrigation_mm	256.2
[6].drainage_1m_mm	75.08
[6].yield_Mg_ha	6.356
[6].transpiration_mm	232.8
[6].soil_evap_mm	145.1
[6].N_leached_kg_ha	27.9
[6].treatment	z100cm_h100_P1.1
[6].soil	S01_CoarseLoam
[7].trig	-100
[7].z	100
[7].P	1.2
[7].ET_mm	699.9
[7].irrigation_mm	235.8
[7].drainage_1m_mm	96.75
[7].yield_Mg_ha	6.511
[7].transpiration_mm	235
[7].soil_evap_mm	148.6
[7].N_leached_kg_ha	23.25
[7].treatment	z100cm_h100_P1.2
[7].soil	S01_CoarseLoam
[8].trig	-100
[8].z	100
[8].P	1.3
[8].ET_mm	698.8
[8].irrigation_mm	214.7
[8].drainage_1m_mm	121.1
[8].yield_Mg_ha	6.605
[8].transpiration_mm	235.1
[8].soil_evap_mm	152.7
[8].N_leached_kg_ha	22.41
[8].treatment	z100cm_h100_P1.3
[8].soil	S01_CoarseLoam
[9].trig	-100
[9].z	100
[9].P	1.4
[9].ET_mm	696.9
[9].irrigation_mm	192
[9].drainage_1m_mm	143.5
[9].yield_Mg_ha	6.62
[9].transpiration_mm	232.6
[9].soil_evap_mm	159.8
[9].N_leached_kg_ha	22.29
[9].treatment	z100cm_h100_P1.4
[9].soil	S01_CoarseLoam
[10].trig	-1200
[10].z	100

quantity	value
[10].P	0.5
[10].ET_mm	663.3
[10].irrigation_mm	426.5
[10].drainage_1m_mm	3.744
[10].yield_Mg_ha	4.54
[10].transpiration_mm	236.1
[10].soil_evap_mm	98.74
[10].N_leached_kg_ha	55.36
[10].treatment	z100cm_h1200_P0.5
[10].soil	S01_CoarseLoam
[11].trig	-1200
[11].z	100
[11].P	0.6
[11].ET_mm	677.2
[11].irrigation_mm	397
[11].drainage_1m_mm	6.238
[11].yield_Mg_ha	5.187
[11].transpiration_mm	239.1
[11].soil_evap_mm	109
[11].N_leached_kg_ha	47.29
[11].treatment	z100cm_h1200_P0.6
[11].soil	S01_CoarseLoam
[12].trig	-1200
[12].z	100
[12].P	0.7
[12].ET_mm	672.7
[12].irrigation_mm	349.1
[12].drainage_1m_mm	10.19
[12].yield_Mg_ha	5.43
[12].transpiration_mm	233
[12].soil_evap_mm	118.4
[12].N_leached_kg_ha	45.91
[12].treatment	z100cm_h1200_P0.7
[12].soil	S01_CoarseLoam
[13].trig	-1200
[13].z	100
[13].P	0.8
[13].ET_mm	671.1
[13].irrigation_mm	304
[13].drainage_1m_mm	14.8
[13].yield_Mg_ha	5.555
[13].transpiration_mm	227.4
[13].soil_evap_mm	129.6
[13].N_leached_kg_ha	50.08
[13].treatment	z100cm_h1200_P0.8
[13].soil	S01_CoarseLoam
[14].trig	-1200
[14].z	100

quantity	value
[14].P	0.9
[14].ET_mm	669
[14].irrigation_mm	257
[14].drainage_1m_mm	18.59
[14].yield_Mg_ha	5.826
[14].transpiration_mm	230.7
[14].soil_evap_mm	137.9
[14].N_leached_kg_ha	41.62
[14].treatment	z100cm_h1200_P0.9
[14].soil	S01_CoarseLoam
[15].trig	-1200
[15].z	100
[15].P	1
[15].ET_mm	673
[15].irrigation_mm	217.7
[15].drainage_1m_mm	21.76
[15].yield_Mg_ha	6.011
[15].transpiration_mm	231.8
[15].soil_evap_mm	149.1
[15].N_leached_kg_ha	37.34
[15].treatment	z100cm_h1200_P1
[15].soil	S01_CoarseLoam
[16].trig	-1200
[16].z	100
[16].P	1.1
[16].ET_mm	662.5
[16].irrigation_mm	166.8
[16].drainage_1m_mm	30.3
[16].yield_Mg_ha	6.184
[16].transpiration_mm	232.1
[16].soil_evap_mm	157.6
[16].N_leached_kg_ha	34.88
[16].treatment	z100cm_h1200_P1.1
[16].soil	S01_CoarseLoam
[17].trig	-1200
[17].z	100
[17].P	1.2
[17].ET_mm	658.8
[17].irrigation_mm	122.5
[17].drainage_1m_mm	35.24
[17].yield_Mg_ha	6.292
[17].transpiration_mm	231.7
[17].soil_evap_mm	169.2
[17].N_leached_kg_ha	32.42
[17].treatment	z100cm_h1200_P1.2
[17].soil	S01_CoarseLoam
[18].trig	-1200
[18].z	100

quantity	value
[18].P	1.3
[18].ET_mm	644.9

e12_pred.json

Source: `Manuscripts/results/e12_pred.json`; 11000 values, first 220 shown. Workbook sheet `R_e12_pred`.

quantity	value
[0].evap	78.21
[0].transp	280.6
[0].et	818.9
[0].irr	667.3
[0].yield	1.619
[0].drain	50.44
[0].nl	33.99
[0].rain	232
[0].other	460.1
[0].store	30
[0].ie	0.2427
[1].evap	93.39
[1].transp	287.3
[1].et	827
[1].irr	629.8
[1].yield	2.614
[1].drain	51.24
[1].nl	27.94
[1].rain	278.4
[1].other	446.4
[1].store	30
[1].ie	0.4151
[2].evap	106.5
[2].transp	301.2
[2].et	840.3
[2].irr	597.2
[2].yield	3.975
[2].drain	51.7
[2].nl	5.26
[2].rain	324.8
[2].other	432.6
[2].store	30
[2].ie	0.6655
[3].evap	118
[3].transp	305.3
[3].et	832.6

quantity	value
[3].irr	545.2
[3].yield	5.087
[3].drain	53.74
[3].nl	0
[3].rain	371.2
[3].other	409.4
[3].store	30
[3].ie	0.9331
[4].evap	134.5
[4].transp	270.9
[4].et	780.5
[4].irr	452.5
[4].yield	6.289
[4].drain	59.66
[4].nl	3.96
[4].rain	417.6
[4].other	375.1
[4].store	30
[4].ie	1.39
[5].evap	144.7
[5].transp	270.2
[5].et	780.7
[5].irr	393.3
[5].yield	6.633
[5].drain	68.84
[5].nl	8.32
[5].rain	464
[5].other	365.8
[5].store	7.743
[5].ie	1.686
[6].evap	154.3
[6].transp	267.2
[6].et	782.3
[6].irr	344.5
[6].yield	6.633
[6].drain	84.62
[6].nl	14.29
[6].rain	510.4
[6].other	360.8
[6].store	-12
[6].ie	1.925
[7].evap	159.1
[7].transp	261.5
[7].et	764.6
[7].irr	297.8
[7].yield	6.633
[7].drain	105
[7].nl	12.99

quantity	value
[7].rain	556.8
[7].other	344
[7].store	-15
[7].ie	2.227
[8].evap	162.1
[8].transp	256.8
[8].et	751
[8].irr	261.5
[8].yield	6.633
[8].drain	128.7
[8].nl	10.94
[8].rain	603.2
[8].other	332.2
[8].store	-15
[8].ie	2.536
[9].evap	163.6
[9].transp	256
[9].et	745.4
[9].irr	235.7
[9].yield	6.633
[9].drain	154.9
[9].nl	6.95
[9].rain	649.6
[9].other	325.9
[9].store	-15
[9].ie	2.814
[10].evap	63.48
[10].transp	209.7
[10].et	596.9
[10].irr	361.5
[10].yield	1.426
[10].drain	11.54
[10].nl	50.76
[10].rain	232
[10].other	323.7
[10].store	-15
[10].ie	0.3944
[11].evap	78.24
[11].transp	221.5
[11].et	622.2
[11].irr	341.1
[11].yield	2.301
[11].drain	12.34
[11].nl	44.71
[11].rain	278.4
[11].other	322.4
[11].store	-15
[11].ie	0.6746

quantity	value
[12].evap	91.82
[12].transp	239
[12].et	650.5
[12].irr	323.5
[12].yield	3.499
[12].drain	12.8
[12].nl	22.03
[12].rain	324.8
[12].other	319.6
[12].store	-15
[12].ie	1.082
[13].evap	105.2
[13].transp	250.6
[13].et	666.7
[13].irr	295.3
[13].yield	4.479
[13].drain	14.84
[13].nl	8.33
[13].rain	371.2
[13].other	310.9
[13].store	-15
[13].ie	1.517
[14].evap	126.1
[14].transp	233.7
[14].et	657
[14].irr	245.1
[14].yield	5.537
[14].drain	20.76
[14].nl	20.73
[14].rain	417.6
[14].other	297.1
[14].store	-15
[14].ie	2.259
[15].evap	136.7
[15].transp	234.9
[15].et	662.1
[15].irr	213
[15].yield	6.345
[15].drain	29.94
[15].nl	25.09
[15].rain	464
[15].other	290.5
[15].store	-15
[15].ie	2.979
[16].evap	146.4
[16].transp	233.3
[16].et	666.3
[16].irr	186.6

quantity	value
[16].yield	6.633
[16].drain	45.72
[16].nl	31.06
[16].rain	510.4
[16].other	286.6
[16].store	-15
[16].ie	3.554
[17].evap	154.6
[17].transp	233.9
[17].et	667
[17].irr	161.3
[17].yield	6.519
[17].drain	66.09
[17].nl	29.76
[17].rain	556.8
[17].other	278.6
[17].store	-15
[17].ie	4.042
[18].evap	161
[18].transp	234.8
[18].et	670
[18].irr	141.7
[18].yield	6.541
[18].drain	89.81
[18].nl	27.71
[18].rain	603.2
[18].other	274.2
[18].store	-15
[18].ie	4.618
[19].evap	165.2
[19].transp	238.1
[19].et	676.3
[19].irr	127.7
[19].yield	6.607
[19].drain	116
[19].nl	23.72
[19].rain	649.6
[19].other	273
[19].store	-15
[19].ie	5.175

S7. Where every table and figure comes from

Item	How it is made	Script(s)	Input and output
Table 1	Descriptive table (definitions, design, parameters or tests); no computation	-	
Figure 1	Figure drawn by the script from the input data	Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/p3_fig01_workflow_v9.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig01_architecture_A4.pdf
Table 2	Table typed from the result file (10 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Table 3	Table typed from the result file (5 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Table 4	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Table 5	Table written by the script (tex/data_table.tex)	Manuscripts/reproducibility/analysis/data_overview.py	Data/Comprehensive_Summary/00_summary_all_simulations
Figure 2	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/data_overview.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/data/fig_data_overview.pdf
Table 6	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 7	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 8	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Table 9	Table typed from the result file (39 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; swa_vars.js; p3_swa_vars.js; p3_swa_fig.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Table 10	Table typed from the result file (15 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; swa_vars.js; p3_swa_vars.js; p3_swa_fig.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Table 11	Table typed from the result file (36 of its numbers found there)	Manuscripts/reproducibility/p3_js/swa_vars.js; Manuscripts/reproducibility/analysis/p3_swa_fig.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p3_swa_vars.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p3_swa_vars.json
Figure 3	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_swa_fig.py; swa_vars.js; p3_swa_vars.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig04_swa_state.pdf

Item	How it is made	Script(s)	Input and output
Figure 4	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_A.jpg
Figure 5	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_B.jpg
Figure 6	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_C.jpg
Table 12	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Results/08_Scripts/novel.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Figure 7	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_sens_figs.py; e1_sens.js; l_sens.js; p3_e1_sens.js; p3_l_sens.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig08_sensitivity.pdf
Table 13	Table typed from the result file (22 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; lib/wsmodel.js; v_surface.js; p3_surface.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p3_ws61.csv
Figure 8	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig03_ws_verification.pdf
Table 14	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; v_border.js; p3_border.js; v_sprinkler.js; run_spr.js; p3_sprinkler.js; spr_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Table 15	Table typed from the result file (15 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; e1_heldout.js; e12.js; e12_pred.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Figure 9	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; make_e12.py; e1_heldout.js; e12.js; e12_pred.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig05_engine1_heldout.pdf
Figure 10	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; e2_closure.js; e2_grid.js; p3_e2_closure.js; p3_e2_grid.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig06_engine2.pdf
Figure 11	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/p3_lit_figs.py; v_drip.js; run_drip.js; p3_drip_swafl.js; drip_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig07_drip.pdf
Figure 12	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_lit_figs.py; run_fao.js; fao_out.js; v_sprinkler.js; run_spr.js; p3_sprinkler.js; spr_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig09_literature.pdf

Item	How it is made	Script(s)	Input and output
Table 16	Table typed from the result file (4 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; v_drip.js; p3_drip_swafl.js; drip_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Figure 13	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_plot_views.png
Figure 14	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_trigger_law.jpg; Manuscripts/figures/shots_jpg/real_mulch_bare.jpg; Manuscripts/figures/shots_jpg/real_root_structure.jpg; Manuscripts/figures/shots_jpg/real_drip_design.jpg
Figure 15	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_field.jpg; Manuscripts/figures/shots_jpg/layer_spac.jpg
Figure 16	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v1.jpg; Manuscripts/figures/shots_jpg/layer_v2.jpg
Figure 17	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_n1.jpg
Figure 18	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v3.jpg; Manuscripts/figures/shots_jpg/layer_v7.jpg
Figure 19	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_r1.jpg
Figure 20	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_e1.jpg; Manuscripts/figures/shots_jpg/layer_v4.jpg
Figure 21	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v6.jpg; Manuscripts/figures/shots_jpg/layer_n2.jpg
Figure 22	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_n3.jpg; Manuscripts/figures/shots_jpg/layer_v5.jpg
Figure 23	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_sid.jpg; Manuscripts/figures/shots_jpg/sid_sec_long.jpg
Figure 24	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/sid_sec_sched.jpg; Manuscripts/figures/shots_jpg/sid_sec_table.jpg
Figure 25	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_drip.jpg; Manuscripts/figures/shots_jpg/drip_sec_layout.jpg
Figure 26	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/drip_sec_lateral.jpg; Manuscripts/figures/shots_jpg/drip_sec_result.jpg
Figure 27	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/drip_sec_table.jpg
Figure 28	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_spr.jpg; Manuscripts/figures/shots_jpg/spr_sec_checks.jpg
Figure 29	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/spr_sec_intake.jpg; Manuscripts/figures/shots_jpg/dash_design.jpg
Figure 30	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/dash_results.jpg; Manuscripts/figures/shots_jpg/dash_charts.jpg
Figure 31	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/dash_messages.jpg
Figure 32	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_field_balance_charts.jpg; Manuscripts/figures/shots_jpg/real_crop_water.jpg

Item	How it is made	Script(s)	Input and output
Figure 33	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_tillage_flux.jpg

References

Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: Guidelines for computing crop water requirements. Technical Report FAO Irrigation and Drainage Paper 56, FAO, Rome, 1998. URL https://www.fao.org/4/x0490e/x0490e00.htm.

M. J. Asher, B. F. W. Croke, A. J. Jakeman, and L. J. M. Peeters. A review of surrogate models and their application to groundwater modeling. *Water Resources Research*, 51:5957–5973, 2015. doi: 10.1002/2015WR016967.

J. E. Christiansen. Irrigation by sprinkling. Technical Report Bulletin 670, University of California Agricultural Experiment Station, Berkeley, 1942.

W. Heber Green and G. A. Ampt. Studies on soil physics: 1. the flow of air and water through soils. *The Journal of Agricultural Science*, 4:1–24, 1911. doi: 10.1017/S0021859600001441.

Jack Keller and David Karmeli. Trickle irrigation design parameters. *Transactions of the ASAE*, 17:678–684, 1974.

John R. Philip. The theory of infiltration: 4. sorptivity and algebraic infiltration equations. *Soil Science*, 84:257–264, 1957.

Saman Razavi, Bryan A. Tolson, and Donald H. Burn. Review of surrogate modeling in water resources. *Water Resources Research*, 48:W07401, 2012. doi: 10.1029/2011WR011527.

Joe T. Ritchie. Model for predicting evaporation from a row crop with incomplete cover. *Water Resources Research*, 8:1204–1213, 1972. doi: 10.1029/WR008i005p01204.

Jiří Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna. Recent developments and applications of the HYDRUS computer software packages. *Vadose Zone Journal*, 15(7), 2016. doi: 10.2136/vzj2016.04.0033.

Pasquale Steduto, Theodore C. Hsiao, Dirk Raes, and Elias Fereres. AquaCrop—the FAO crop model to simulate yield response to water: I. concepts and underlying principles. *Agronomy Journal*, 101:426–437, 2009. doi: 10.2134/agronj2008.0139s.

Wynn R. Walker and Gaylord V. Skogerboe. *Surface Irrigation: Theory and Practice*. Prentice-Hall, Englewood Cliffs, NJ, 1987.